



American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

Trip Diary-Travel Management & Tour Suggestion System

A Software Engineering Project Submitted

By

Semester: Summer 24-25		Section:	Group Number:	
SL	Student Name	Student ID	Contribution (CO3+CO4)	Individual Marks
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5				

The project will be evaluated for the following Course Outcomes

CO3 (PO-g-1) <i>Select appropriate software engineering models, project management roles and their associated skills for the complex software engineering project and evaluate the sustainability of developed software, taking into consideration the societal and environmental aspects</i>	Total Marks	
Selection of Software Engineering Models: Process model selection and presents sufficient evidence to support argument for the model selection	[5 Marks]	
Role identification and Responsibility Allocation: Well-planned project with proper role identification and responsibility allocation in the project management activities	[5Marks]	
Formatting and Submission: Submission, Defense, Completeness, Spelling, grammar, and Organization of the Project report	[5Marks]	
CO4 (PO-k-1) <i>Apply engineering management principles and economic decision making to develop software engineering project management plan.</i>	Total Marks	
Project WBS and Testcases: Relevant WBS (project task list) and testcases for the proposed project are stated properly.	[5Marks]	
Effort Estimation and Scheduling: Project estimation was described using proper effort estimation or schedules based on available project resources	[5Marks]	
Risk Management: Sufficient and appropriate risks are identified, analyzed, and properly categorized or prioritized.	[5Marks]	

Project Proposal-Trip Diary-Travel Management & Tour Suggestion System

1. PROJECT PROPOSAL

1.1 BACKGROUND OF THE PROBLEM

Trip Diary is a smart and friendly travel management and planning platform made for people who love to explore, document, and share their journeys. It's designed to make traveling easier, more organized, and more enjoyable, whether you're looking back on past adventures or planning your next trip. With a secure and simple account system, users can sign up, log in, and start building their own digital travel diaries. They can record details about places they've visited, upload photos and videos, track expenses, and write notes about their favourite memories, all in one convenient place. Planning new trips is just as easy. Trip Diary helps users create personalized travel plans, estimate travel costs, and keep track of bookings or tickets, so everything they need is right at their fingertips.

One of the most exciting parts of Trip Diary is the Smart Tour Suggestion System, which learns from each user's travel history and recommends new destinations that match their interests. It's like having a travel guide who knows your style. It won't suggest places you've already been, and you can even choose whether you want to explore destinations within your own country or go international. Beyond personal trip planning, Trip Diary also brings people together through its Community Tour Event feature. Users can create and share their own group travel events, invite others to join, or browse events hosted by fellow travellers. It's a fun and interactive way to meet new people, share experiences, and explore the world together.

Behind the scenes, Trip Diary includes an Admin Panel to keep the platform safe, organized, and up to date. Admins can manage users, locations, and shared content, making sure the system runs smoothly and securely for everyone. In short, Trip Diary isn't just another travel app, it's a digital travel companion that helps users plan smarter, travel better, and relive their favourite memories anytime. By combining journaling, smart recommendations, and a community-driven experience, Trip Diary turns every journey into something more meaningful and unforgettable.

1.2 SOLUTION TO THE PROBLEM AND PROCESS MODEL SELECTION

1.2.1 Project Scopes

This project aims to provide travellers with a complete platform to plan, organize, and manage their trips efficiently. It helps travellers with destination details, travel schedules, cost estimates, and booking assistance, thereby enhancing user convenience and travel experience.

1.2.2 Project Features

- **Destination Highlights:** Basic info and attractions of places like Panam City, Sylhet, Nepal, and Sikkim.
- **Travel Itinerary:** Day-wise trip planning with clear schedules.
- **Cost Estimator:** Average travel and accommodation cost.
- **Tour Packages:** Budget, mid-range, and luxury options for different travelers.
- **Photo & Info Gallery:** Visuals with short descriptions of tourist spots.
- **Cultural Insights:** Short notes on local traditions, food, and heritage.
- **User-Friendly Layout:** Clean interface for quick access to information, etc.

1.2.3 User Story Card/Table

Table 01. User Story Card

As a/an	I want to	So that	Acceptance Criteria
Guest User	Browse public tour events	I can explore travel opportunities	Can view event title, destination, and limited info without login
Guest User	View destination suggestions	I can get inspired to travel	System shows demo suggestions with limited personalization
Guest User	Sign up/log in	I can access the full features	Successful login or registration shows profile dashboard
Registered User	Add past trips	I can log and revisit my travel history	User provides destination, date, notes, and optional media
Registered User	Plan upcoming trips	I can organize my future travel plans	User adds destination, dates, and preferences
Registered User	Upload travel media	I can document and share my trips	Uploading photos/videos with trip details is successful
Registered User	View and update profile	I can keep my account info current	Editable profile view with update confirmation
Registered User	Get travel suggestions	I can discover new places to visit	Suggestions are based on travel history and user preference (domestic/international)
Registered User	Create public tour events	I can invite others to join my trip	Event form submitted with location, date, cost, and visibility
Registered User	Enrolling in public tour events	I can join others' tours	Enrollment success confirmation shown
Registered User	View cost estimations	I can estimate travel budget	System shows cost estimate based on input like destination, duration, and travel type

Admin	Approve/monitor public events	I can control what is visible to others	Event goes live only after admin approval
Admin	Manage destination database	I can ensure accurate and updated travel info	Admin can add/edit/delete destinations
Admin	Monitor uploaded media	I can keep the content appropriate	Uploaded media flagged or approved
Admin	Manage user accounts	I can control access to the system	Admin can view, block, or delete any user account
Admin	Oversee system performance	I can ensure smooth functioning	Admin dashboard shows system logs, user activity, and reports
System	Suggest new destinations	I can recommend places not yet visited	System compares travel history with available database to generate suggestions

1.2.4 User Story Board/ Trello Board

10/9/25, 3:37 PM

Trip Diary | Trello

Backlog

To Do

- Choose domestic or international travel type for registered users
- Approve or monitor public tour events for admin
- Manage the destination database for admin
- Monitor uploaded photos and videos for admin
- Manage user accounts (view, delete, block) for admin
- Monitor Performance for admin

In Progress

- Enroll in available tour events for registered users
- View average cost estimations for trips for registered users

<https://trello.com/b/ms93fqzG/trip-diary>

1/3

Figure 01. Trello Board Page-01

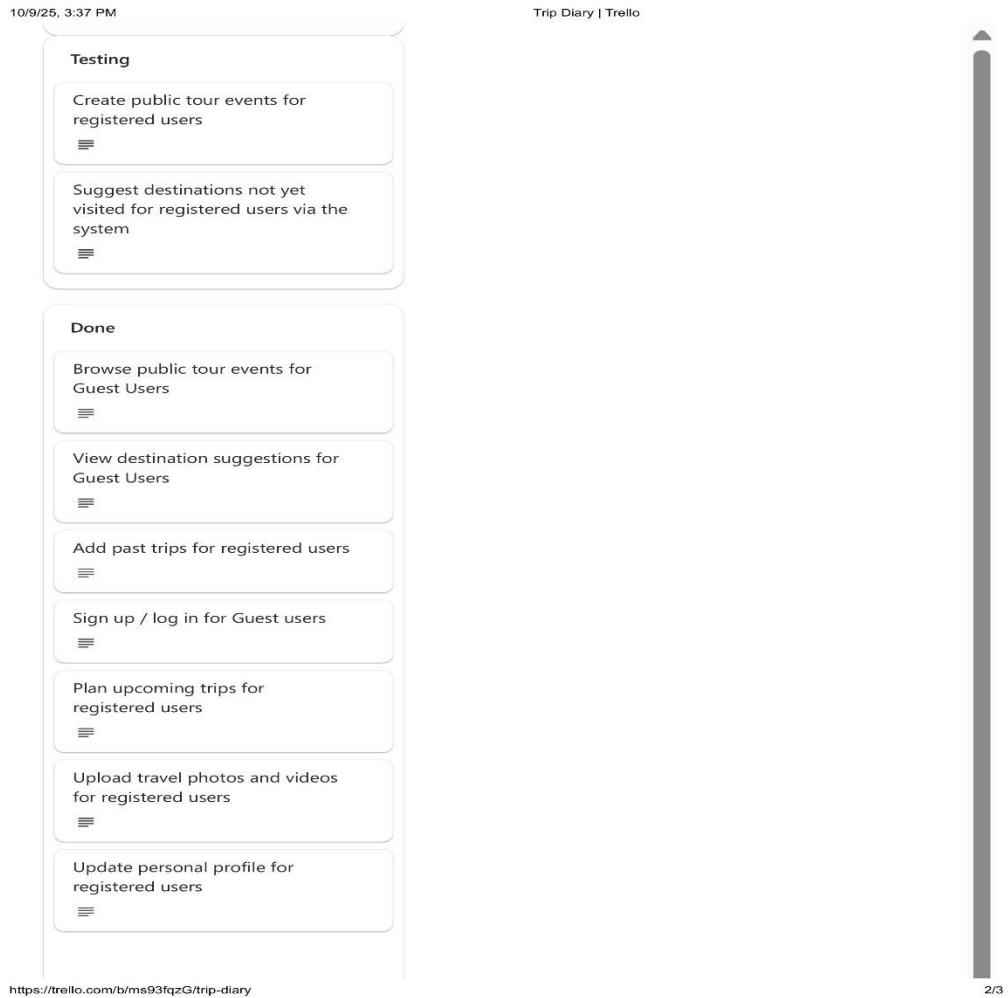


Figure 02. Trello Board Page-02

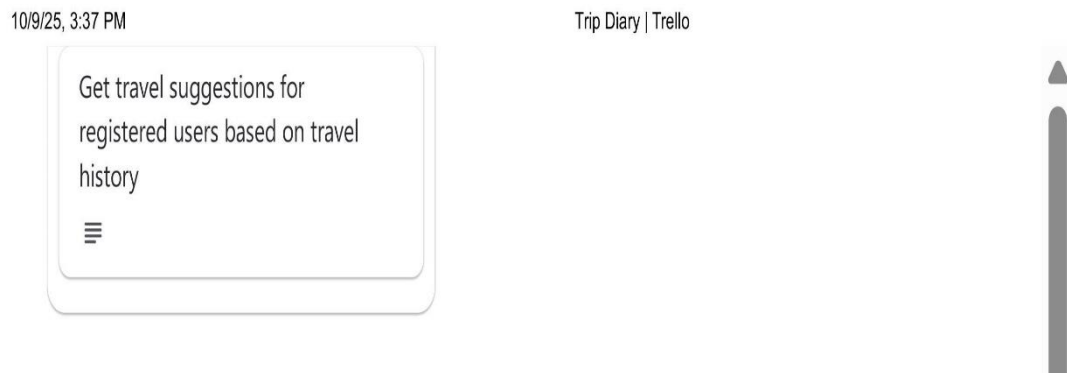


Figure 03. Trello Board Page-03

1.2.5 The existing software solutions that are available to solve the problem mentioned

1. Pratra EPIC
2. Travelpro Trip Builder/ Itinerary Management
3. Trawax Itinerary Builder & Travel Portal
4. FlightsLogic Trip Builder
5. Travefy

1.2.6 Software Development Process Model

Scrum Process Model: We chose Scrum because it allows for regular feedback and updates by dividing work into smaller sprints, which is more suitable for changing needs like travel projects. XP primarily focuses on coding practices, but Scrum is more effective in overall team management and feature delivery.

1.2.7 Analysis of the Project Environment

Nature of Requirements:

- Functional Requirements: Browsing events, adding trips, enrolling in tours, and administrative control.
- Non-functional Requirements: Performance, security, usability, scalability, and reliability are key, given that users will store personal travel data and media.

Stability of Requirements:

Likely to Change:

1. Travel suggestions and recommendation algorithms may evolve as user preferences and behavior patterns become clearer.
2. UI/UX elements may be refined based on user feedback.
3. Social features, such as the ability to share trips or media, could expand based on insights gained from user engagement.

Relatively Stable:

1. Core features, including user registration, trip logging, event creation, and cost estimation, are fundamental and are unlikely to change significantly.
2. Administrative functions for monitoring, approving events, and managing content are stable, as they are important for the maintenance of the system.

Project Environment Factors:

1. Dynamic environment: User expectations, travel trends, and personalization needs may change over time.

2. Multiple types of users input: Inputs from users, admins, and business requirements will guide development.
3. Collaboration and feedback dependency: Continuous user testing and admin feedback are crucial to refine travel recommendations, UI/UX, and event functionalities.

1.2.8 Implications for Scrum

The project fits Scrum well because:

1. We know the SCRUM Process has a Scrum Master. The Scrum Master makes sure the project follows Scrum rules, values, and practices, and that the work moves forward as planned. So, our project will proceed in a planned way.
2. The customer helps with tasks related to the product backlog items for the system that is being built or improved. We defined a Project backlog for our Project. The backlog Practice technique is good for the Scrum Process. So, Scrum Process is more suitable for our project than other Processes.
3. Requirements are partially developed, and frequent sprints allow for incremental improvements.
4. We should have a daily Progress meeting for tracking our project Progress.
5. Regular sprint reviews provide opportunities to validate features like travel suggestions, media uploads, and cost estimations.

The Scrum model is well-suited for the Trip Diary-Travel Management & Tour Suggestion project because it highlights collaboration, adaptability, and iterative progress.

Team Size: Scrum works best with small to medium-sized teams. We have 4 group members to complete this project. This setup is common for projects like this and makes sure that each team member can handle their own part, such as frontend work, backend coding, database setup, or testing.

Communication: Daily scrum meetings and sprint reviews encourage open communication, where every member share progress, updates and upcoming work.

Task Coordination: The product backlog ensures that all requirements are clearly prioritized. The sprint backlog then breaks them into manageable tasks. This makes coordination easier, as members know what to deliver in each sprint.

So, we can say that the Scrum-based solution is suitable to meet the business objectives of our project.

Flexibility: Some features may change over time. Scrum helps adjust and improve them regularly based on user feedback.

Step-by-Step Delivery: Important features can be finished and used early, so the system is useful even before it's fully done.

User-Focused: Regular sprint reviews let travelers and admins test the system and share feedback, ensuring the product meets their needs.

Lower Risk: Problems can be spotted and fixed early, reducing the chance of big failures later.

1.2.9 Flexibility of the Scrum Model

The Scrum model is very flexible and adaptable, which makes it a great choice for the Trip Diary project.

1. **Changes in Scope:** Since Scrum works in short, repeatable cycles known as sprints, it's easy to handle changes in project scope. New ideas or features can be added to the product backlog, prioritized, and then included in future sprints without affecting current work.
2. **Changes in Technology:** Scrum supports step-by-step improvements, so adopting new technologies becomes smoother. Once a new tool or feature is reviewed during sprint planning, it can be introduced in the next sprint.
3. **Changes in User Requirements:** As users ask for more personalized travel suggestions or new features like social sharing, the team can quickly respond. Feedback from sprint reviews helps refine and re-prioritize user stories, allowing updates to be delivered in upcoming iterations without starting over.

1.2.10 Deep Insight & Creative Solution

Problem: Travelers often face difficulties due to the lack of a single, reliable source of travel information. They have to browse multiple websites for destinations, budgeting, and itinerary planning which makes the whole process confusing, time-consuming, and sometimes frustrating.

Creative Solution: Our project introduces a smart travel management system that brings all travel-related needs together on one platform. It also offers:

1. Day-by-day itinerary planning, helping travelers organize their trips clearly and efficiently.
2. Automatic cost estimation in the local currency for better transparency in budgeting.
3. Personalized tour package recommendations (budget, mid-range, or luxury) made with user preferences.
4. Built-in flexibility to easily add new destinations, features, or technologies in the future.

This approach transforms a stressful planning process into a simple, guided experience that helps travelers save time, reduce confusion, and make well-informed decisions.

1.2.11 Targeted Group of Users & Benefits

Target Group of Users: The primary target users are travelers and tourists (both domestic and international) who want to discover destinations like Panam City, Sylhet, and Sikkim. Secondary users include tour operators and travel agencies who can use the system to design and share customized packages.

Users Benefits:

1. Travelers will benefit from a one-stop platform that provides destination info, day-wise itineraries, cost estimation, package comparisons, saving time, reducing confusion, and making trips more budget-friendly.
2. Tour operators will benefit from presenting ready-made plans, which will improve customer trust and attract more clients through a transparent and organized system.

1.2.12 Contribution to Scientific Results

This project helps improve the fields of software engineering and information systems by using the Scrum model to solve real problems in the tourism sector. The main contributions are:

1. Finding the Real Needs: The project carefully studies what travelers need like destination information, trip planning, and cost estimation to make travel easier.
2. Using the Scrum Process: The Scrum model is used to show how working in sprints makes the project more flexible and able to adapt to changes quickly.
3. Combining Knowledge: It shows how software can bring together cultural details, travel costs, and planning tools into one simple platform for users.
4. Clear and Practical Results: The project works as a real example of how technology can fill information gaps, make travel planning faster, and inspire future research in smart tourism systems.

1.2.13 Evidence for Model Selection

The Scrum process model is a perfect choice for a travel management system because it matches the fast-changing nature of travel projects. These projects often require quick updates like adding new destinations, changing prices, or launching seasonal offers. Scrum's short and repeatable sprints make it easy to handle such changes without disrupting the whole system.

- Flexibility: Each sprint delivers a working part of the system such as destination details, cost estimation, or travel packages which helps the project move forward smoothly.
- User Feedback: After every sprint, users can test the new features and share their opinions. This is especially useful for travel apps, where user needs and trends often change.
- Risk Reduction: Since Scrum releases updates in small steps, it helps avoid large failures that can happen when everything is delivered at once.

- **Team Collaboration:** Scrum clearly defines roles Product Owner, Scrum Master, and Development Team which makes teamwork more organized and efficient, especially in a project that involves multiple areas like design, data, and travel management.

In short, Scrum is the most flexible, user-focused, and practical method for developing the proposed travel management system.

1.2.14 How Scrum Tackles Risks and Uncertainties

Scrum helps manage risks at every stage of development by breaking the project into smaller, manageable parts and allowing continuous feedback and improvement.

1. Requirement Stage (Backlog Creation)

- **Risk:** Unclear or shifting requirements.
- **Scrum Solution:** All requirements are listed in the Product Backlog and prioritized. This makes it easy to add or update items anytime without disturbing the whole project.

2. Planning Stage (Sprint Planning)

- **Risk:** Setting unrealistic goals or taking on too much work.
- **Scrum Solution:** The team selects only achievable tasks for each sprint, ensuring the workload is clear and manageable.

3. Development Stage (Sprints)

- **Risk:** Technical problems or system failures.
- **Scrum Solution:** Work is done in small, continuous parts (increments), allowing the team to detect and fix issues early before they grow into bigger problems.

4. Testing & Review Stage (Sprint Review)

- **Risk:** Features might not meet user needs.
- **Scrum Solution:** After each sprint, stakeholders test and review the results. Their feedback helps the team stay aligned with user expectations.

5. Monitoring Stage (Scrum meeting)

- **Risk:** Miscommunication within the team.
- **Scrum Solution:** Short daily meetings keep everyone updated, highlight challenges, and ensure smooth coordination.

6. Delivery Stage (Increment Release)

- **Risk:** Discovering major issues too late.
- **Scrum Solution:** Since working features are released regularly, any issues are caught early, and previous parts of the system continue to function properly.

1.2.15 Scrum & Project Schedule

1. **Sprint-Based Timeline:** The project is organized into short, fixed-length sprints. Every sprint result in a working module (such as destination info, cost estimator, itinerary

planner). This approach makes sure that progress is clear and can be measured against the timeline.

2. **Timely Delivery:** Since each sprint gives a usable piece, the project remains on track for its deadlines. Even if the entire project isn't finished, a functional version can still be delivered on schedule.
3. **Flexibility with Deadlines:** If unexpected changes pop up, like new destinations or updated cost data, Scrum allows for adjustments to the backlog without throwing off the entire schedule. Workload is handled realistically because tasks are divided into sprint goals.
4. **Continuous Monitoring:** Daily meetings and sprint reviews help catch delays early and adjust tasks, ensuring everything stays on track for timely completion.

1.2.16 Justification for Choosing Scrum

Scrum is the most suitable model for this project compared to alternatives like XP (Extreme Programming) or FDD (Feature Driven Development). The reasons are:

1. **Adaptability to Change:** Travel management systems need constant updates such as new destinations, pricing changes, and seasonal offers. Scrum's short and repeatable sprints make it easy to include these updates without affecting the whole system. In contrast, XP and FDD are less flexible when it comes to changing requirements.
2. **User-Centered Development:** Scrum focuses strongly on user feedback through sprint reviews and backlog refinement. This ensures the system grows according to what users want. XP mainly emphasizes technical coding practices, while FDD focuses more on early design rather than ongoing user input.
3. **Better Time Management:** Scrum delivers small, usable parts of the system within short time periods. This keeps progress steady and reduces the risk of big delays or failures. XP works fast but may lack long-term planning, while FDD takes more time upfront for detailed modeling.
4. **Scalability and Team Collaboration:** Scrum is ideal for larger or multidisciplinary teams, thanks to its clear roles such as Scrum Master, Product Owner, and Development Team. XP is better for very small, highly skilled groups, and FDD can be too complex for smaller projects.

1.3 PROJECT ROLE IDENTIFICATION AND RESPONSIBILITIES

1.3.1 Main Roles in the Project

1. **Product Owner:** Umme Fahmida Sultana
 - Sets the vision and goals for the project.
 - Oversees the Product Backlog, which includes features, requirements, and priorities.
 - Make sure the final product aligns with what users expect.

2. **Scrum Master:** Eyamin Khan Emon
 - Guides the team through Scrum practices like sprint planning, daily meetings, and reviews.
 - Works to eliminate any obstacles that might hinder the team's progress.
 - Ensures that communication flows smoothly and that the workflow is efficient.
3. **Project Stakeholders:** Marin Rahman & Rashed Faruque
 - Comprises potential users, travel agencies, or other organizations.
 - Offers feedback and helps validate the project's progress.

1.3.2 Development Roles

1. **Developers or Programmers:** Eyamin Khan Emon & Marin Rahman
 - Bring system features to life, working on the frontend, backend, databases, and integrations.
2. **UI/UX Designer:** All Team
 - Crafts a user-friendly interface and improves the overall customer experience.
3. **Tester:** All Team
 - Test each deliverable from the sprint for errors, performance issues, and usability.
 - Guarantees quality before the product is released.
4. **Database Administrator:** Rashed Faruque & Marin Rahman
 - Manages travel-related data, including destinations, costs, packages, and itineraries.
 - Make sure that the data is secure and can scale as needed.
5. **System Analyst:** Umme Fahmida Sultana & Rashed Faruque
 - Converts requirements into technical solutions.
 - Aligns the system design with the project's goals.

1.3.3 Responsibilities by Project Stage

1. **Requirements Gathering:**
 - **Product Owner:** Gathers requirements from stakeholders and prioritizes features in the Product Backlog.
 - **Scrum Master:** Acts as a bridge, Maintaining communication between stakeholders and the development team.
 - **System Analyst:** Takes those requirements and turns them into clear, technical specifications.
 - **Users:** Share their insights on needs, preferences, and the problems that need solving.
2. **Design:**
 - **UI/UX Designer:** Crafts frames, layouts, and user-friendly interfaces that enhance the user experience.
 - **System Analyst:** Outlines the system architecture, database models, and overall workflow.
 - **Developers:** Offer feedback on the design's feasibility and suggest any improvements.
 - **Product Owner:** Checks to ensure the design aligns with the initial requirements.

3. Implementation:

- **Programmers:** Write the code for both the frontend and backend, as well as system features.
- **Database Administrator:** Constructs and manages the database structure.
- **Scrum Master:** Works to eliminate any blockers and keeps the team focused on sprint goals.
- **Product Owner:** Clarifies any requirements and decides whether to accept or reject completed features.

4. Testing:

- **Tester:** Conducts unit testing, integration testing, and user acceptance testing to ensure quality.
- **Developers:** Address any bugs that come up during the testing phase.
- **Product Owner:** Confirms that the features meet the acceptance criteria.
- **Stakeholders:** May take part in user acceptance testing to provide additional feedback.

5. Deployment:

- **Developers & DBA:** Handle the deployment of the system to a live server or environment.
- **Scrum Master:** Manages the deployment schedule and ensures everything goes off without any drawbacks.
- **Product Owner:** Gives the final approval for the release after confirming that all backlog items for the sprint are completed.
- **Users:** Begin using the system and share their feedback for future Sprints.

1.3.4 Specifying roles are responsible for decision making, quality assurance, and resource management.

1. Decision Making:

- **Product Owner:** Takes charge of project priorities, backlog items, and feature approvals.
- **Scrum Master:** Focuses on improving processes, facilitating sprints, and resolving any conflicts that arise.
- **Development Team:** Collaborates on technical solutions, coding standards, and design decisions.

2. Quality Assurance (QA):

- **Tester:** Makes sure features are bug-free and thoroughly tested before they go live.
- **Developers:** Gently keep an eye on code quality, unit tests, and integration stability.
- **Scrum Master:** Ensures that Scrum practices are adhered to, maintaining overall quality.
- **Product Owner:** Gives the green light on deliverables based on acceptance criteria.

3. Resource Management:

- **Scrum Master:** Oversees team productivity, clears obstacles, and distributes tasks fairly.
- **Product Owner:** Manages business resources, including time, budget, and requirements.
- **Project Stakeholders:** Provide funding and any external resources needed.

- **Development Team:** Handles technical resources like tools, code repositories, servers, and databases.

1.3.5 Distribution of responsibilities among the team members and Justification of the allocation based on skills and expertise

1. **Product Owner:** Umme Fahmida Sultana

Responsibilities:

- Shape the product vision and set clear goals.
- Keep the Product Backlog organized and prioritized.
- Make sure every feature meets user needs, like travel logging, destination suggestions, and cost estimation.
- Offer clarifications to the development team during sprints.
- Give the green light on completed user stories at the end of each sprint.

Justification: The Product Owner should have a solid understanding of travel systems and what users prioritize in their experience. They need the authority to decide which features matter most to users.

Skills: strong communication, business analysis, and understanding user requirements.

2. **Scrum Master:** Eyamin Khan Emon

Responsibilities:

- Lead Scrum ceremonies, including Sprint Planning, Daily meetings, Sprint Review, and drawbacks.
- Eliminate any obstacles that hinder the team's progress.
- Ensure the team conforms to Scrum practices and upholds Agile principles.
- Monitor team progress and keep an eye on sprint performance.

Justification: This role requires excellent team management and facilitation skills, rather than deep technical expertise. The Scrum Master serves as a vital link between the Product Owner and the Development Team.

Skills: Adept problem-solving, leadership, conflict resolution, and a solid understanding of Scrum methodology.

3. **Development team:** All Team

Responsibilities:

- **Backend Developer:** Responsible for building the database, server, and APIs that handle trips, events, and suggestions.
- **Frontend Developer:** Tasked with designing the user interface for browsing events, logging trips, and creating dashboards.
- **Tester:** Conducts thorough testing to ensure the system operates flawlessly.
- **Deployment Engineer:** Manages the deployment of the application and ensures it runs smoothly over time.

Justification: Each team member focuses on tasks that align with their skills and expertise. This approach promotes efficient development and results in high-quality features and clearly defined roles help avoid confusion and improve collaboration within the team.

1.3.6 Justification of the allocation based on skills and expertise

1. **Domain-Expertise Alignment:** The Product Owner focuses on what users need and what's most important, leaving the technical details to the experts.
2. **Technical Efficiency:** Developers are matched with tasks that suit their strongest skills. Backend work goes to backend professionals, while frontend tasks are handled by UI specialists.
3. **Clear Accountability:** Every role comes with clear responsibilities for specific tasks, which helps prevent any overlap or confusion.
4. **Agility & Collaboration:** The Scrum Master plays a key role in communication and teamwork between technical and non-technical members, allowing the team to adapt swiftly to any changes.

2. SOFTWARE REQUIREMENTS SPECIFICATIONS (SRS) / PRODUCT REQUIREMENTS DOCUMENT (PRD)

Table 02. PRD Table

Template Name	Trip Diary- Travel management & Tour Suggestion System
Goals & Objective	The primary goal of the Trip Diary system is to develop a user-friendly travel management platform that allows users to plan, record, and share their travel experiences while receiving personalized trip suggestions and cost estimations. Objectives: •Log past trips and plan upcoming travel with helpful notes, dates, and media uploads •Create or join public tour events to travel in groups •Get personalized travel suggestions based on their history and preferences •View estimated costs for planned trips using destination, duration, and travel type •Browse public tour events and destinations (for guests, with limited access) •Update their personal profile information anytime •Admins can manage destinations, user accounts, and moderate uploaded content
Client Information	
Product Type	Windows Application / Android Application
Domain	Web2
Access Layers	Guest User, User, admin, System
Design Required	Yes
Feature 01: App Overview for Guest Users & Auth Feature	

Phase	Item No	Specification [Role]	Screen Definition	User Story	Acceptance Criteria	UI Screen / Wireframe	Tags
Phase 1	1	Browse public tour events [Guest Users]		As a Guest User, I want to browse public tour events so that I can explore travel opportunities.			Public Tour Event
phase 1	2	View Destination [Guest Users]		As a Guest User, I want to view destination suggestions so that I can get inspired to travel.			Destinations
phase 2	3	Signup & Login [Guest Users]		As a Guest User, I want to sign up or log in so that I can access full features.			Signup & Login
Phase 2	4	Forgot password [admin] [User]		As an Admin & User, I want to reset my Password so that I can Login the System.			Forgot Password
Feature 02: Dashboard							
Phase	Item No	Specification [Role]	Screen Definition	User Story	Acceptance Criteria	UI Screen / Wireframe	Tags
Phase 2	5	Admin Interface		As an Admin, I want to access a centralized admin interface so that I can efficiently manage users, destinations, public events, uploaded media, and system performance.			Interface
phase 3	6	User Interface		As a registered user, I want a clear and user-friendly interface so that I can easily navigate the platform, plan trips, log travel history, and interact with tour events without confusion.			Interface
phase 3	7	User Interface [Admin][Users]		As a system user (Admin/User), I want a clear and user-friendly interface so that I can easily access and perform my specific tasks without confusion.			Menu or Dashboard
Feature 03: Registered User Activities							
Phase	Item No	Specification [Role]	Screen Definition	User Story	Acceptance Criteria	UI Screen / Wireframe	
Phase 4	7	Add Past Trips [user]		As a Registered User, I want to add past trips so that I can log and revisit my travel history.			Past Trips
phase 4	8	Plan Upcoming Trips [User]		As a Registered User, I want to plan upcoming trips so that I can organize my future travel plans.			Upcoming Trips
phase 4	9	Upload Media [User]		As a Registered User, I want to upload travel media so that I can document and share my trips.			Media
Phase 4	10	Update Personal Profile [User]		As a Registered User, I want to update my profile so that my account info stays current.			Personal Profile
Phase 4	11	Travel Suggestion based on Travel history [User]		As a Registered User, I want to get travel suggestions so that I can discover new places.			Travel Suggestion
Phase 4	12	Create Public Tour events		As a Registered User, I want to create public tour events so that I can invite others.			Create Events
Phase 4	13	Choose domestic or International Travel type		As a registered user, I want to choose between domestic or international travel so that I receive personalized suggestions and cost estimates.			Travel Type

Phase 5	14	Enroll Tour Event		As a Registered User, I want to enroll in public tour events so that I can join others' tours.			Enroll Event
Phase 6	15	average cost estimation		As a Registered User, I want to view cost estimations so that I can plan my budget.			Cost
Feature 04: Payment Feature							
Phase	Item No	Specification [Role]	Screen Definition	User Story	Acceptance Criteria	UI Screen / Wireframe	Tags
Phase 7	16	Payment for Enrolled Tour Event [User]		As a registered user, I want to make secure payments for tour events or travel bookings so that I can confirm my participation without any hassle.			Payment
Feature 05: Admin Activities							
Phase	Item No	Specification [Role]	Screen Definition	User Story	Acceptance Criteria	UI Screen / Wireframe	Tags
Phase 3	17	Approve/Monitor public Tour Event [Admin]		As an Admin, I want to approve or monitor public events so that I can control visible content.	Monitor Event		signup
phase 3	18	Manage Destination Database [Admin]		As an Admin, I want to manage the destination database so that info is accurate.	Destinations		signin
phase 3	19	Monitor Uploaded Media's [Admin]		As an Admin, I want to monitor media uploads so that content stays appropriate.	Monitor Media		forgot password
phase 3	20	Manage User Accounts [Admin]		As an Admin, I want to manage user accounts so that I can control system access.	Manage User		
phase 3	21	Monitor System's Performance [Admin]		As an Admin, I want to oversee system performance so that I can ensure smooth operation.	Performance		
Feature 06: System Activities							
Phase	Item No	Specification [Role]	Screen Definition	User Story	Acceptance Criteria	UI Screen / Wireframe	Tags
Phase 1	22	Suggest Destinations [System]		As a System, I want to suggest destinations not visited by the user so that they discover new places.	Suggest Destination		signup

2.1 FUNCTIONAL REQUIREMENT

2.1.1 Major functionalities of Trip Diary

- User Authentication:** This feature lets users sign up, log in, and securely manage their accounts. It ensures that only registered users can access personalized features tailored just for them.
- Manage Past Trips:** Users can store all the details of their past adventures, including destinations, dates, notes, and even media like photos and videos. This helps users look back on and organize their travel history.
- Trip Planning:** Users can plan their future getaways by specifying destinations, dates, and personal preferences. It offers a neat overview of their upcoming travel schedules.
- Destination Suggestions:** The system recommends exciting new destinations based on users' travel history and preferences, whether they're looking to explore domestically or internationally. It cleverly avoids suggesting places users have already been, encouraging them to discover new spots.
- Public Tour Event Management:** Users can create, view, and join public tour events with admin approval, all events are assess to ensure they're appropriate before being visible to others.
- Cost Estimation:** Users can estimate their travel budgets by entering details like destination, travel type, and trip duration. This feature supports smart financial planning for future adventures.

7. **Media Upload and Management:** Users can upload photos and videos from their trips, creating a visual diary of their experiences. Admins keep an eye on the content to ensure everything is appropriate.
8. **Admin Control & Monitoring:** Admins can approve events, manage destinations, monitor media, oversee users, and ensure the system runs smoothly. This helps maintain system integrity, data quality, and safe interactions within the community.
9. **Smart Suggestion Engine:** The system automatically suggests new destinations that users haven't visited yet. It provides personalized travel recommendations based on their historical data.

2.1.2 Core Services, Operations, and Features

1. User Services

- Sign Up / Log In.
- Profile Management.
- Password Management.

2. Trip Management

- Log Past Trips.
- Plan Upcoming Trips.
- View Trip History.

3. Destination & Travel Suggestions

- Personalized Destination Recommendations.
- Demo Suggestions for Guests.

4. Public Tour Events

- Create Events.
- Enroll in Events.
- View Events.

5. Travel Cost Estimation

- Estimate Trip Costs.

6. Media Management

- Upload Photos/Videos.
- View Media.

7. Admin & Monitoring

- Event Approval.
- Destination Database Management.
- User Account Management.
- Media Monitoring.

8. System Automated Features

- Smart Destination Suggestions.
- Notifications / Alerts.

2.1.3 How each function supports the overall objectives of the project

1. User Authentication

Function: Sign up, log in, and manage accounts securely.

Support: Make sure users have personalized access to log their trips, plan their journeys, and get tailored recommendations. Safeguards user data and privacy, which is crucial for building trust in a travel platform.

2. Manage Past Trips

Function: Users can store details of their trips, including where they went, when, notes, and photos.

Support: Allows users to keep a record of their travel history, which is key for reflecting on past experiences and planning future adventures. Offers data for personalized suggestions, boosting user engagement.

3. Trip Planning

Function: Users can map out future trips by adding destinations, dates, and preferences.

Support: Helps organize travel schedules, making it easier for users to manage their trips. Supports community events and cost estimates, adding an interactive element to travel planning.

4. Destination Suggestions

Function: The system suggests new places to visit based on users' travel history and preferences.

Support: Encourages users to discover new destinations, aligning to make travel exploration easy and personalized. Avoids suggesting places users have already visited, improving overall satisfaction.

5. Public Tour Event Management

Function: Users can create, view, and sign up for public tour events.

Support: Fosters a community-driven travel experience, allowing users to connect, collaborate, and begin on trips together. Incorporates social features into the travel management system, boosting user engagement.

6. Cost Estimation

Function: Users can estimate trip costs based on destination, travel type, and duration.

Support: Aids users in budgeting and financial planning, making the platform practical and user-friendly. Encourages smarter travel choices.

7. Media Upload and Management

Function: Users can easily upload photos and videos from their trips.

Support: This feature improves the travel journaling experience, making the platform more interactive and visually engaging. It also creates a vibrant collection of travel memories and event content.

8. Admin Control & Monitoring

Function: The admin is responsible for approving events, managing destinations, monitoring media, and ensuring the system runs smoothly.

Supports: This role is crucial for maintaining data integrity, safety, and the quality of user-generated content. It helps to uphold system reliability and fosters community trust.

9. Smart Suggestion Engine

Function: This feature suggests destinations that the user hasn't visited yet.

Support: It offers personalized travel discovery, boosting user engagement and satisfaction. Additionally, it automates valuable services that increase the platform's functionality without requiring manual input.

In summary, every function plays a vital role in Trip Diary's mission to help users log, plan, discover, share, and enjoy their travel experiences, all while ensuring security, reliability, and community involvement.

2.1.4 Workflows that represent how a user will complete a specific task through the system

Workflow 1: User Login

- The user opens the Trip Diary system.
- They select the Login option.
- Next, they enter their username or email along with their password.
- The system checks the credentials against the database.
- If everything checks out, then the user is taken to their dashboard.
- If something's off, then the system displays an error message and tells them to try again.

Workflow 2: Logging a Past Trip

- The user logs into the system.
- They choose the “**Log Past Trip**” option.
- They fill in the trip details, including destination, travel dates, and any notes.
- They can also upload photos or videos if they want.
- After that, they click Save.
- The system saves the trip in the user's travel history.

Workflow 3: Planning a Future Trip

- The user logs into the system.
- They select the “**Plan New Trip**” option.
- They enter the destination, intended dates, and their preferences.
- They can check the cost estimation for the planned trip.
- Finally, they save the plan.

Workflow 4: Destination Suggestions

- The user logs into the system.
- They navigate to the “**Trip Suggestions**” page.
- The system reviews the user’s travel history and preferences, whether domestic or international.
- It then generates a list of new destinations that the user hasn’t visited yet.
- The user can view these suggestions and may choose to add one to their future plans.

Workflow 5: Creating a Public Tour Event

- The user logs into the system.
- They select the “**Create Event**” option.
- They enter the event details, including destination, date, capacity, and description.
- They submitted the event request.
- An admin reviews and approves the event.
- Once approved, the event becomes visible to all users.

Workflow 6: Joining a Tour Event

- The user logs into the system.
- They browse through public events.
- They select an event that they like.
- They click “**Enroll Event**”.
- The system enrolls the user and updates the event participant list.

Workflow 7: Cost Estimation

- The user logs into the system.
- They head over to the “**Estimate Cost**” option.
- Next, they fill in the details: destination, travel type (bus, air, train), and how long the trip will be.
- The system pulls up historical data and calculates the numbers to give an estimated budget.
- The user checks out the estimated cost.

Workflow 8: Admin Moderation

- The admin logs into the system through the Admin Panel.
- They check new event creation requests, user reports, and any uploaded content.
- The admin either approves or rejects requests and removes any inappropriate content if needed.
- They also update the destination database whenever new places are added

2.1.5 Requirements using user stories with Acceptance Criteria that determine when a functional requirement is successfully implemented.

1. As a Guest User, I want to browse public tour events so I can explore travel opportunities.
Acceptance Criteria: I can see the event title, destination, and some basic info without needing to log in.
2. As a Guest User, I want to check out demo destination suggestions to spark my travel inspiration.
Acceptance Criteria: The system provides limited, non-personalized suggestions.
3. As a Registered User, I want to keep track of my past trips so I can look back on my travel history
Acceptance Criteria: I can save details like destination, date, notes, and optional media.
4. As a Registered User, I want to plan my upcoming trips to keep my future travels organized.
Acceptance Criteria: I can successfully add my destination, dates, and preferences.
5. As a Registered User, I want to upload travel media to document my adventures.
Acceptance Criteria: I can upload photos and videos along with my trip details.
6. As a Registered User, I want to receive personalized travel suggestions to discover new places.
Acceptance Criteria: The system suggests destinations based on my travel history and preferences (both domestic and international).
7. As a Registered User, I want to create public tour events to invite others to join me.
Acceptance Criteria: The event is created with location, date, cost, and visibility, pending admin approval.
8. As a Registered User, I want to sign up for public tour events so I can travel with others.
Acceptance Criteria: I receive confirmation of my enrollment, and the participant list is updated.
9. As a Registered User, I want to estimate travel costs to help me budget for my trips.
Acceptance Criteria: The system provides cost estimates based on inputs like destination, duration, and travel type.
10. As an Admin, I want to approve and monitor events to ensure quality and safety.
Acceptance Criteria: Events only go live after receiving approval.
11. As an Admin, I want to manage destinations to keep the database accurate and up to date.
Acceptance Criteria: I can add, edit, or delete destinations as needed.
12. As an Admin, I want to oversee uploads to maintain quality control.
Acceptance Criteria: Media should be flagged or approved before it goes public.
13. As an Admin, I want to manage user accounts so that I can control access.
Acceptance Criteria: The admin should have the ability to block, delete, or view accounts.

2.1.6 Acceptance Criteria that determine when a functional requirement is successfully implemented

- I can see the event title, destination, and some basic info without needing to log in.
- The system provides limited, non-personalized suggestions.
- I can save details like destination, date, notes, and optional media.

- I can successfully add my destination, dates, and preferences.
- I can upload photos and videos along with my trip details.
- The system suggests destinations based on my travel history and preferences (both domestic and international).
- The event is created with location, date, cost, and visibility, pending admin approval.
- I receive confirmation of my enrollment, and the participant list is updated.
- System provides cost estimates based on inputs like destination, duration, and travel type.
- Events only go live after receiving approval.
- I can add, edit, or delete destinations as needed.
- Media flagged or approved before public view.
- Admin can block, delete, or view accounts.

2.2 NON-FUNCTIONAL REQUIREMENT

The Trip Diary system needs to do more than just its main tasks; it also has to deliver top-notch quality in performance, security, usability, and scalability. Here is the classification of the non-functional requirements we expect:

Performance

- The system should load the main dashboard in under 5 seconds when the network is running smoothly.
- It must support at least 200 users at the same time without slowing down.
- Cost estimation and destination suggestions should pop up within 5 to 7 seconds.

Reliability

- The system should aim for 99% uptime each month.
- If something goes wrong, like a server crash, it should recover within 5 minutes using backups.
- Event enrollment and trip logging should be foolproof, ensuring no data is lost even if a disruption happens during the process.

Integrity & Security

- All sensitive information, like passwords and user details, must be stored securely in an encrypted format.
- We should implement two-factor authentication for a safer login experience.
- Role-based access control will make sure that only admins can approve events, manage destinations, or moderate content.
- The system needs to observe basic privacy standards, ensuring personal information isn't exposed to unauthorized users.

Usability

- The interface should be straightforward, user-friendly, and optimized for mobile devices.
- We should offer multi-language support to cater to a diverse user base.
- The system must include guides and tooltips to assist new users.

Maintainability

- The code should be modular and follow standard naming conventions to make debugging easier.
- Updates and bug fixes should be rolled out without disrupting ongoing services, utilizing version control.
- Documentation for APIs and the database schema must be kept up to date for future developers.

Scalability

- The system needs to handle an increase from 200 to 1000 users at the same time without requiring a lot of configuration changes.
- The database should efficiently manage the growing number of travel history records, media uploads, and event data.
- To accommodate more users, we can implement cloud deployment and load-balancing strategies to scale horizontally.

3. PROJECT ESTIMATION AND SCHEDULING

3.1 Effort & Cost Estimation

3.1.1 Scope of the Project

The Trip Diary Travel Management System is designed to offer a variety of features for different users:

- **Guest Users**, they can easily browse events, check out demo suggestions, and sign up or log in.
- **Registered Users** will have the ability to log their past trips, plan future adventures, upload photos, view personalized suggestions, estimate costs, create or join public events, and manage their profiles.
- **Admins** will have the tools to approve and monitor events, manage the destination database, oversee uploaded media, handle user accounts, and keep an eye on system performance.

This project encloses the development of a web-based application with a responsive user interface, backend services, and seamless database integration.

3.1.2 Lines of Code (LOC) estimation by predicting the size of the system and converting it into effort using productivity rates

We assumed that the system is a Small to Medium Web Application with multiple features.

Estimated size of the system: 12,700 LOC

Project Effort Estimation:

$$E = \left[\frac{LOC \times B^{0.333}}{P} \right]^3 \times \left(\frac{1}{t^4} \right)$$

Estimated size of the system: 12,700 LOC

Productivity Factors, B=1.12(assumed for medium business app or web application)

Productivity parameter, P=600LOC/pm

Estimated time, t=4 months

Now,

$$E = \left[\frac{LOC \times B^{0.333}}{P} \right]^3 \times \left(\frac{1}{t^4} \right)$$

$$E = \left[\frac{12700 \times 1.12^{0.333}}{600} \right]^3 \times \left(\frac{1}{4^4} \right)$$

$$E = 41.48 \approx 42 \text{ Person/Months(Answer)}$$

3.1.3 COCOMO model by selecting the appropriate project mode and calculating effort, development time, and staffing

- **Person-Months needed for project (Effort Estimation),**

$$\text{Effort} = \text{PM} = \text{Coefficient}_{\langle \text{Effort Factor} \rangle} \times \left(\frac{SLOC}{1000} \right)^P$$

- **Duration time in months for the project,**

$$DM = 2.50 \times (PM)^T$$

- Required Number of people,

$$ST = \frac{PM}{DM}$$

We assumed,

Our project type is a semi-detached project.

So,

Coefficient_{<Effort Factor>}=3.0

P=1.12

T=0.35

SLOC=12,700

Now,

Person-Months Needed for Project,

$$\text{Effort} = PM = \text{Coefficient}_{\text{Effort Factor}} \times \left(\frac{SLOC}{1000}\right)^P$$

$$\text{Effort} = PM = 3.0 \times \left(\frac{12700}{1000}\right)^{1.12}$$

$$\text{Effort} = PM = 51.68 \text{ (Answer)}$$

Development Time,

$$DM = 2.50 \times (PM)^T$$

$$DM = 2.50 \times (51.68)^{0.35}$$

$$DM = 9.945 \text{ Week – days (Answer)}$$

Required Number of people,

$$ST = \frac{PM}{DM}$$

$$ST = \frac{51.68}{9.945}$$

ST = 5.196 $\approx$ (5 – 6) Person (Answer)

Results from all three estimation methods

After analysing the estimations-

- Expected effort = 45-48 person-months.
- Expected team size = 5-6 developers.
- Expected schedule = 9-10 months.

3.1.4 Assumptions and possible variation in results

Based on our Project, the estimated Level of Effort (LOC) can vary by about $\pm 15\%$, depending on how complex the project is.

When we talk about productivity rates, we're assuming a team with average skills. If we have a team with more or less experience, that could change the amount of effort needed.

The COCOMO model is based on the Semi-detached type, but if the complexity rises to embedded, then we can expect the effort to increase as well.

As for estimation uncertainty, it's normal to see a variation of 10-25% during the early stages of planning.

3.2 PROJECT SCHEDULING

3.2.1 Work Breakdown Structure

The project is broken down into smaller, more manageable tasks, each with its own set of responsibilities:

Requirement Analysis & Planning: This involves gathering requirements, defining the project scope, and preparing the Software Requirements Specification.

Responsibilities: Project Manager, Business Analyst.

System Design: Here, we focus on database design, system architecture, and creating UI/UX wireframes.

Responsibilities: System Architect, UI/UX Designer.

Implementation: This stage includes backend coding for authentication, trip management, and the event system, as well as frontend coding for the user interface, dashboards, and forms.

Responsibilities: Developers (both Frontend & Backend).

Testing: We conduct manual testing, unit testing, integration testing, and system testing to ensure everything works smoothly.

Responsibilities: QA/Test Engineers and Developers.

Deployment & Maintenance: Finally, we deploy the system, monitor its performance, and fix any bugs that arise.

Responsibilities: Maintenance Team.

3.2.2 Effort allocation across phases, such as analysis/design, coding, and testing, using the 40–20–40 rule

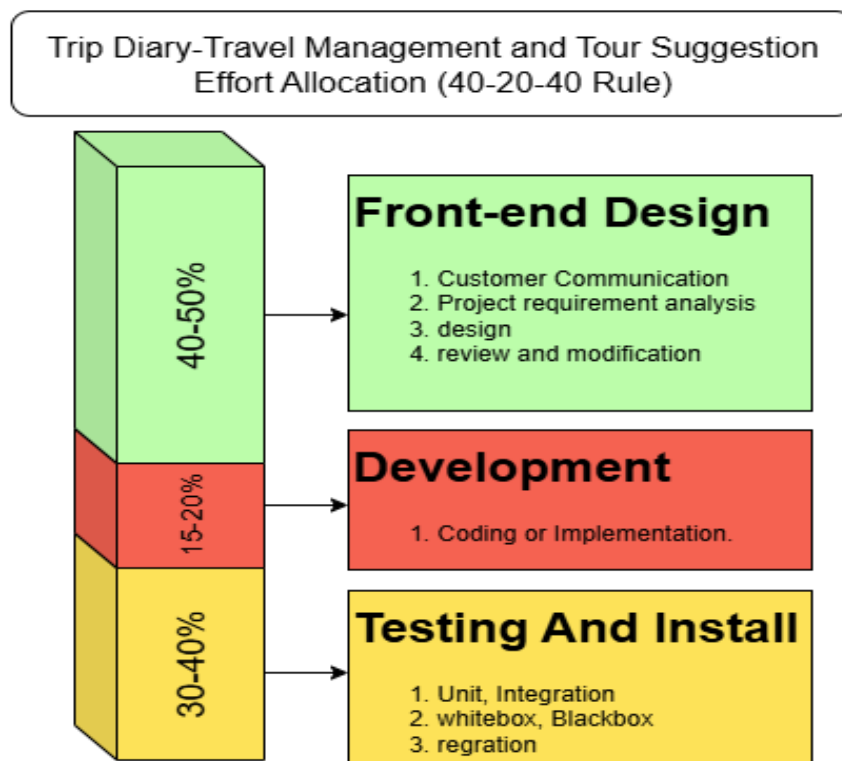


Figure 04. Effort allocation (40-20-40 Rule)

Using the 40–20–40 Rule:

- Analysis & Design (Front-end Design): 40/50% - (4/5 months).
- Implementation: 20% - 2 months (frontend & backend).
- Testing & Deployment: 40% - 4/5 months (testing, bug fixing, deployment).

3.2.3 A schedule with milestones and deadlines, represented in a timeline or Gantt chart (Manually)

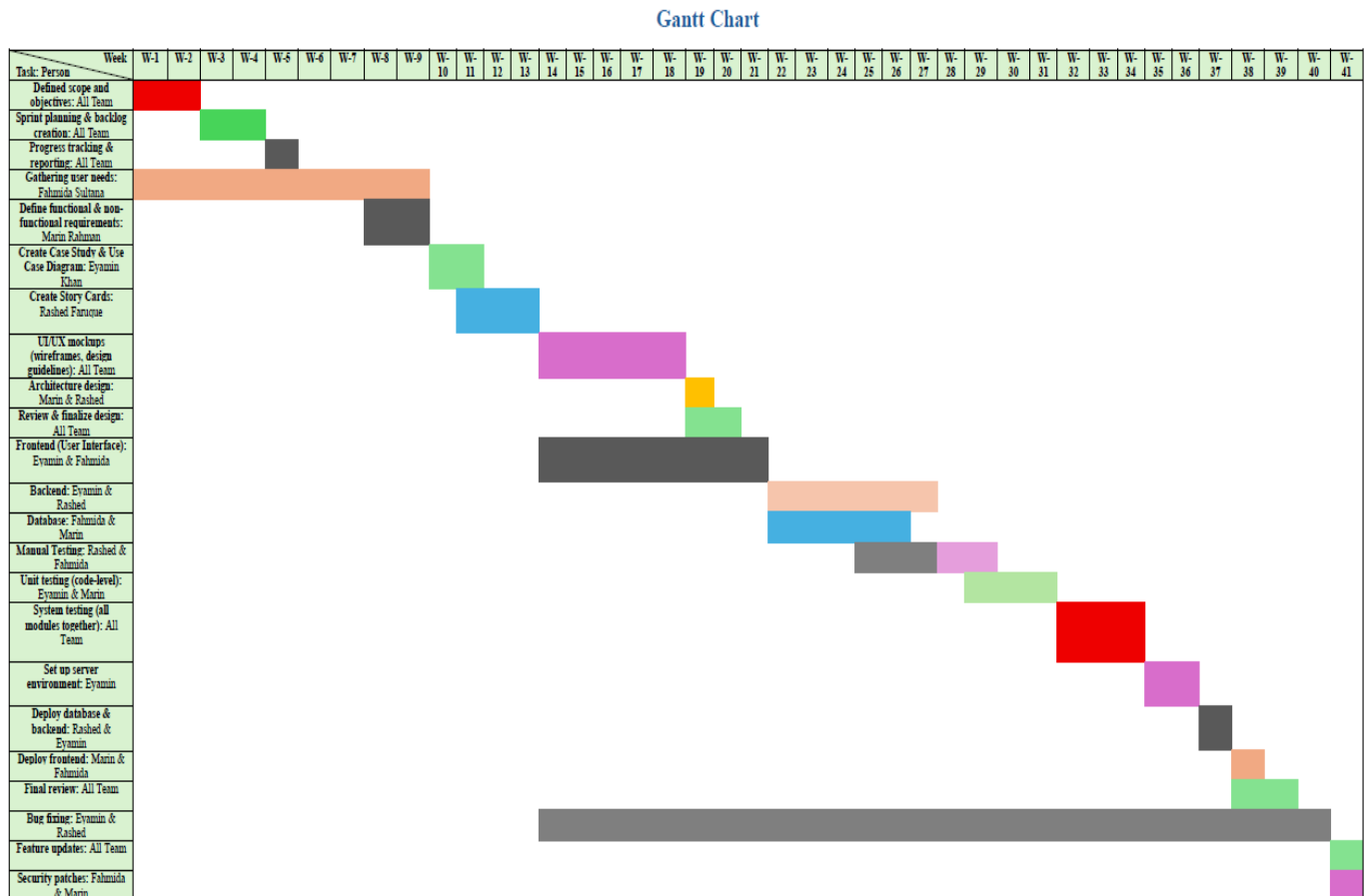


Figure 05. Gantt Chart

A schedule with milestones and deadlines, represented in a timeline or Gantt chart (Using Notion's features)

Link:

https://www.notion.so/27ab245fec3480609092c904f9b4f25b?v=27ab245fec3481a89a8a000c46cf343d&source=copy_link

3.2.4 Major Deliverable tied to Milestones and Outputs

1. Project Initiation

We begin by planning and assigning roles.

- Deliverable: A straightforward project plan that outlines responsibilities for each team member.
- Milestone: The team is aligned, and the project is officially underway.

2. Understanding What We're Building

We gather all the requirements from users and administrative needs.

- Deliverable: A complete Software Requirements Specification.
- Milestone: Everyone is aligned on what the system should achieve.

3. Designing the System

We create the overall architecture and design the database.

- Deliverables: A system design document, diagrams, and the database schema.
- Milestone: The system blueprint is ready for the developers.

4. First Development Phase (Core Features)

We focus on building the essentials: user login, trip logging, planning, and cost estimation.

- Deliverables: A working prototype that includes authentication and trip management features.
- Milestone: The first demo version of the system is ready for testing.

5. Second Development Phase (Advanced Features)

We add some exciting elements: destination suggestions, event management, media uploads, and admin controls.

- Deliverables: Fully functional modules for suggestions, events, and an admin dashboard.
- Milestone: All system features are complete.

6. Testing the System

We thoroughly check every component: units, integrations, and the entire system, fixing any bugs along the way.

- Deliverables: Test reports and a stable version of the system.
- Milestone: The system passes all tests and is ready for release.

7. Deployment

We set up the live environment and launched the system for users.

- Deliverables: A running system, a deployment guide, and a user manual.
- Milestone: Users can access and utilize the platform live.

8. Maintenance & Support

We keep focused on performance, resolving issues, and roll out updates as needed.

- Deliverables: Regular patches, performance reports, and ongoing support.
- Milestone: The system is reliable and up to date.

3.2.5 Tracking Progress

To keep the project on the right path, the team will:

Hold regular meetings: We'll have short weekly stand-up meetings to go over what's been completed, what's currently in the works, and any challenges we might be facing. This way, everyone stays on the same page, and we can catch any issues early on.

Review milestones: At the end of each major phase like, requirements, design, development, testing, and deployment. we'll take a moment to ensure that everything we planned is complete and functioning as it should. These reviews serve as important checkpoints to confirm we're making progress.

Catch the schedule: We'll frequently update the project timeline (Gantt chart) to compare what we planned to do with what we've completed. If we notice any delays, we'll adjust prevent bigger problems down the line.

3.2.6 Earned Value Analysis (EVA) with measures

Table 03. Effort Data Table

Tasks	Planned Effort	Actual Effort
1	15.0	15.0
2	12.5	14.0
3	14.5	13.0
4	8.5	8.0
5	8.5	8.5
6	10.0	8.5
7	4.0	4.5
8	9.5	9.0
9	3.5	3.5
10	12.5	9.0
11	11.5	10.5
12	17.5	15.5
13	15.5	10.0
14	7.5	7.0
15	12.0	11.0
16	6.5	7.0

17	8.5	10.5
18	12.0	10.5
19	17.0	15.0
20	9.5	10.5
21	15.0	15.0
22	18.5	18.0
23	4.0	6.0
24	8.5	8.0
25	8.0	9.0
26	12.5	12.0
27	13.5	14.0
28	14.0	16.0
29	10.5	9.0
30	6.0	6.5
31	15.0	14.5
32	4.5	5.0
33	5.0	7.0
34	7.5	5.5
35	16.0	14.5
36	11.5	12.0
37	10.5	10.0
38	13.0	12.5
39	6.0	7.0
40	9.5	10.5
41	15.5	14.0
42	10.0	12.0
43	7.0	8.5
44	8.0	7.5
45	12.5	12.0
46	16.5	15.0
47	9.5	9.5
48	12.0	12.5
49	16.0	16.5
50	13.5	12.5
51	10.0	10.5
52	14.5	14.0
53	8.0	8.5
54	8.5	7.0
55	15.5	16.0
56	12.0	13.0
57	10.0	11.5
58	11.5	10.0
59	16.5	15.0
60	13.0	12.0
61	10.5	12.5
62	14.5	13.0
63	7.5	9.0
64	10.5	11.5
65	12.5	12.0

66	15.0	14.5
67	10.0	10.0
68	11.0	9.0
69	16.5	15.5
70	12.5	13.5
71	9.0	9.5
72	14.0	13.5
73	11.5	11.0
74	16.5	16.0
75	12.5	11.5
76	9.0	9.5
77	14.0	14.0
78	11.5	11.5
79	15.0	15.0
80	12.5	12.0
81	14.5	14.0
82	8.5	8.5
83	8.5	8.0
84	10.0	10.5
85	4.0	5.5
86	9.5	10.5
87	3.5	4.0
88	12.5	12.0
89	11.5	11.5
90	17.5	15.0
91	15.5	15.0
92	7.5	7.0
93	12.0	12.0
94	6.5	6.5
95	8.5	9.5
96	12.0	12.5
97	17.0	17.5
98	9.5	7.5
99	15.0	13.0
100	18.5	14.0
101	4.0	3.5
102	8.5	7.2
103	8.0	6.5
104	12.5	5.5
105	11.0	11.5
106	7.0	7.5
107	10.5	10.0
108	10.5	11.5
109	15.0	12.0
110	10.5	10.0
111	15.0	13.5
112	18.0	16.5
113	6.0	6.5
114	8.0	8.0

115	9.0	9.5
116	12.0	12.5
117	14.0	14.0
118	16.0	16.0
119	9.0	8.5
120	6.5	7.0
121	14.5	14.0
122	5.0	5.5
123	7.0	6.0
124	5.5	5.5
125	12.0	11.5
126	6.5	6.0
127	8.5	7.0
128	12.0	10.5
129	17.0	13.5
130	9.5	11.0
131	15.0	13.5
132	18.5	17.5
133	4.0	5.5
134	8.5	8.0
135	8.0	8.5
136	12.5	12.5
137	11.0	11.5
138	7.0	6.5
139	10.5	10.0
140	10.5	10.5
141	15.0	14.5
142	10.5	10.0
143	15.0	15.0
144	18.0	18.5
145	6.0	5.5
146	8.0	8.5
147	9.0	12.5
148	14.0	13.5
149	16.0	
150	9.0	
151	6.5	
152	14.5	
153	5.0	
154	7.0	
155	5.5	
156	15.0	
157	18.0	
158	6.0	
159	8.0	
160	9.0	
161	14.0	
162	16.0	
163	9.0	

164	6.5
165	14.5
166	5.0
167	7.0
168	5.5
169	8.5
170	14.0
171	13.0
172	13.5

Here,

Budgeted cost of work performed, **BCWP=1642.5**

Budgeted cost of work scheduled, **BCWS=1888.5**

Actual cost of work performed, **ACWP=1597.5**

Total tasks=183

Effort Estimated=2005-person day

Now,

Budget at Completion,

$$BAC = \sum (BCWS_k)$$

Schedule Performance Index,

$$SPI = \frac{BCWP}{BCWS}$$

Schedule Variance,

$$SV = BCWP - BCWS$$

Percent Schedule for Completion,

$$\text{Percent-Schedule-for-Completion} = \frac{BCWS}{BAC}$$

Percent Complete,

$$\text{Percent-Complete} = \frac{BCWP}{BAC}$$

Cost Performance Index,

$$CPI = \frac{BCWP}{ACWP}$$

Cost V,

$$CV = BCWP - ACWP$$

Calculations...

$$BAC = \sum (BCWS_k)$$

$$BAC = 2005.00 \text{ (Answer)}$$

$$SPI = \frac{BCWP}{BCWS}$$

$$SPI = \frac{1642.5}{1888.5}$$

$$SPI = 0.869 \text{ (Answer)}$$

$$SV = BCWP - BCWS$$

$$SV = 1642.5 - 1888.5$$

$$SV = -246 \text{ Person-day (Answer)}$$

$$\text{Percent-Schedule-for-Completion} = \frac{BCWS}{BAC}$$

$$\text{Percent-Schedule-for-Completion} = \frac{1888.5}{2005.00}$$

$$\text{Percent-Schedule-for-Completion} = 94.2\% \text{ (Answer)}$$

94.2 % of work scheduled to be done at this time.

$$\text{Percent-Complete} = \frac{BCWP}{BAC}$$

$$\text{Percent-Complete} = \frac{1642.5}{2005.00}$$

$$\text{Percent-Complete} = 81.9\% \text{ (Answer)}$$

81.9 % of work completed at this time.

$$CPI = \frac{BCWP}{ACWP}$$

$$CPI = \frac{1642.5}{1597.5}$$

CPI=1.02 (Answer)

$$CV = BCWP - ACWP$$

$$CV = 1642.5 - 1597.5$$

CV=45 Person-day (Answer)

3.2.7 Recognizing and Managing Delays

In the world of real projects, delays can arise for a various of reasons. Here are some of the most common Delays:

Unrealistic deadlines: When timelines are too tight, the quality of work can suffer, or tasks might be rushed and left unfinished.

Plan: Set realistic deadlines based on how much effort is needed and include some buffer time for those unexpected hiccups.

Changing requirements: If the features or goals keep shifting during development, it can really slow things down.

Plan: Lock in the core requirements early on and manage any new changes through a structured process.

Risks: Technical challenges, dependency failures, or even outside factors can make difficulties in progress.

Plan: Spot potential risks ahead of time and have backup strategies ready to go.

Technical or human issues: Bugs, skill gaps, or a team member suddenly being unavailable can lead to delays.

Plan: Make sure team members are cross-trained and ready to provide quick support when issues arise.

Miscommunication: If information isn't shared clearly, tasks can be misunderstood or duplicated.

Plan: Keep communication clear and consistent through regular stand-ups, shared documents, and milestone reviews.

4. SOFTWARE DESIGN

4.1 Diagrams

4.1.1 System design for Trip Diary-Travel Management & Tour Suggestion System

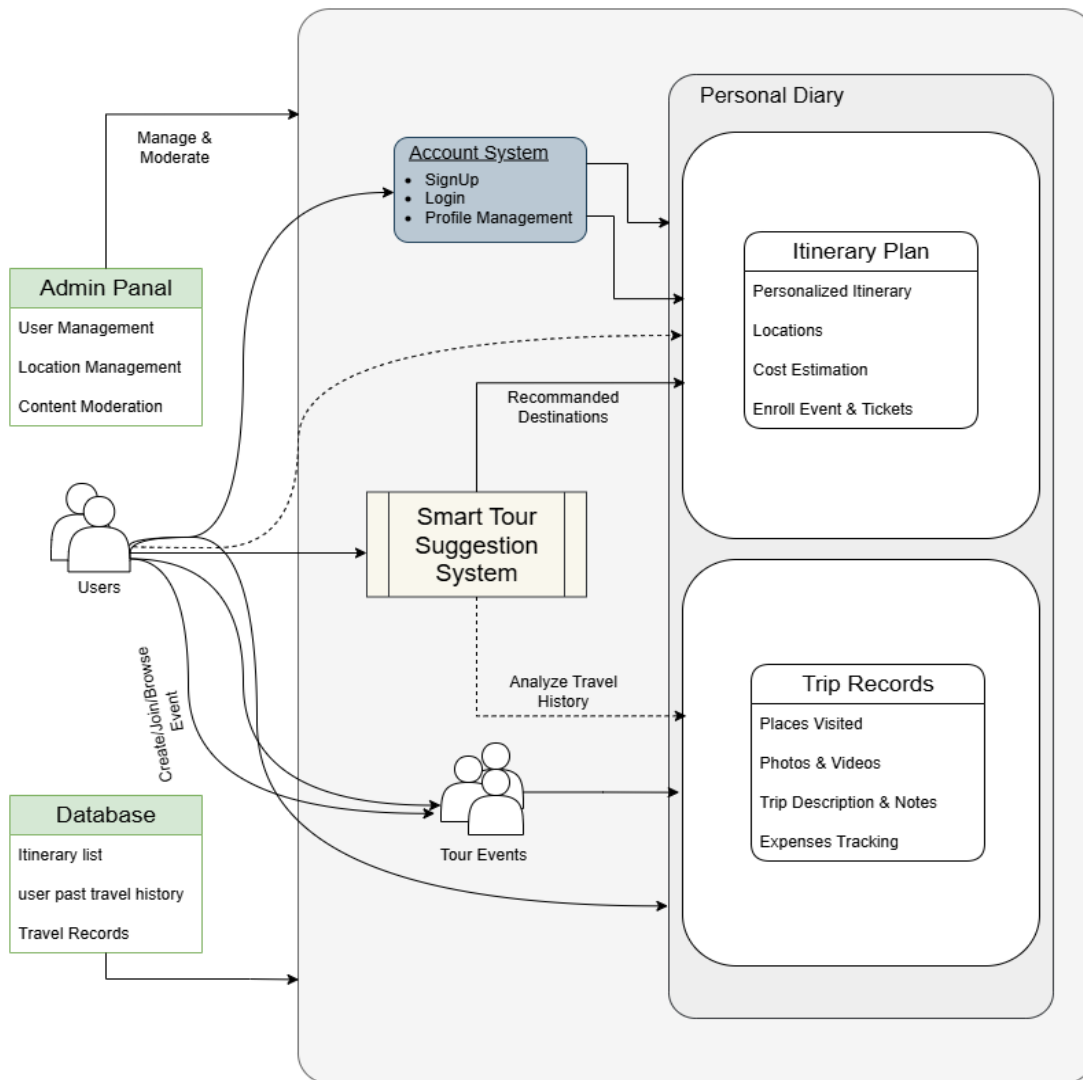


Figure 06. System Diagram

4.1.2 Use Case Diagram by first defining all users (actors) and their roles

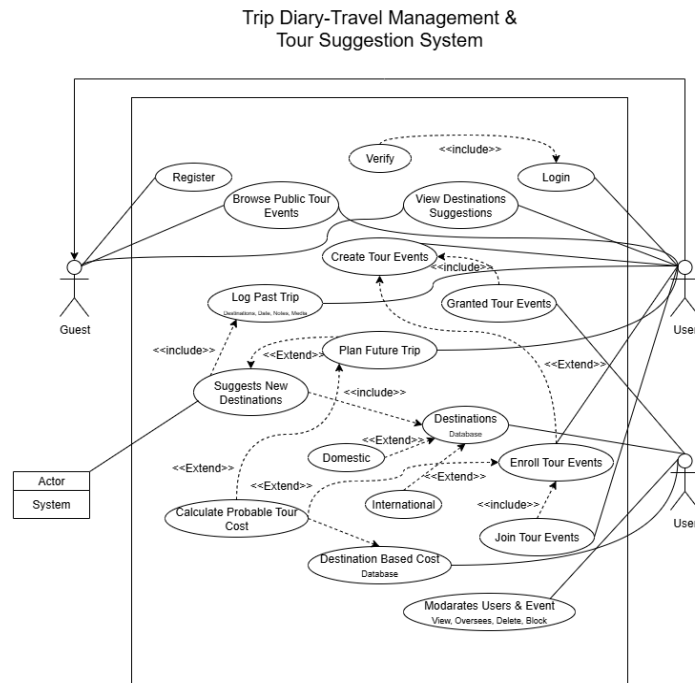


Figure 07. Use case Diagram

4.1.3 Class Diagram for Trip Diary-Travel Management & Tour Suggestion System

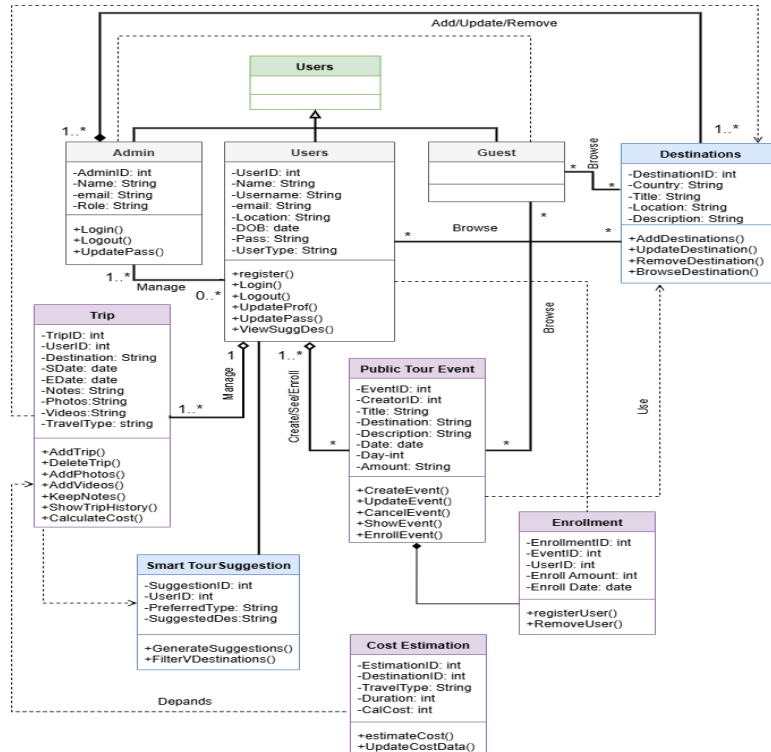


Figure 08. Class Diagram

4.1.4 Activity Diagram that visually represents the workflow of a system

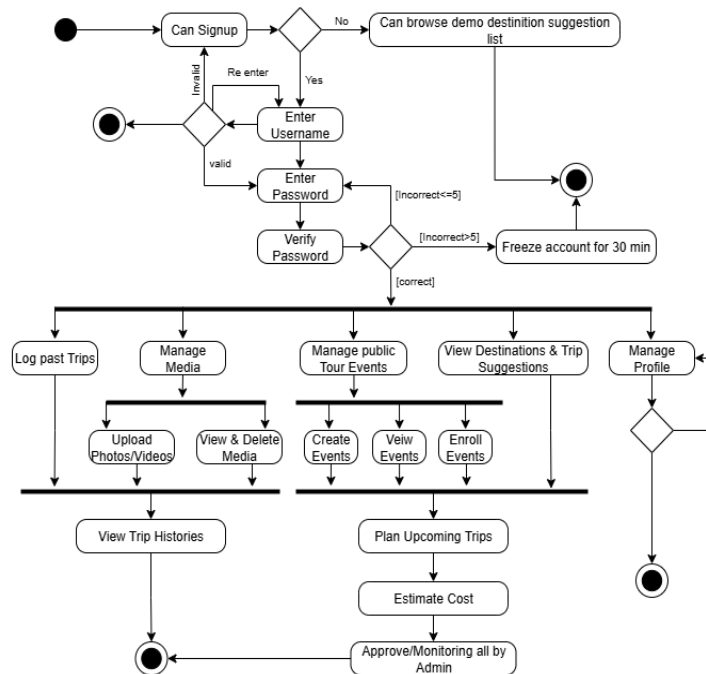


Figure 09. Activity Diagram

4.1.5 A Data Flow Diagram (DFD) starting with the Context Diagram (system as one process with external entities)

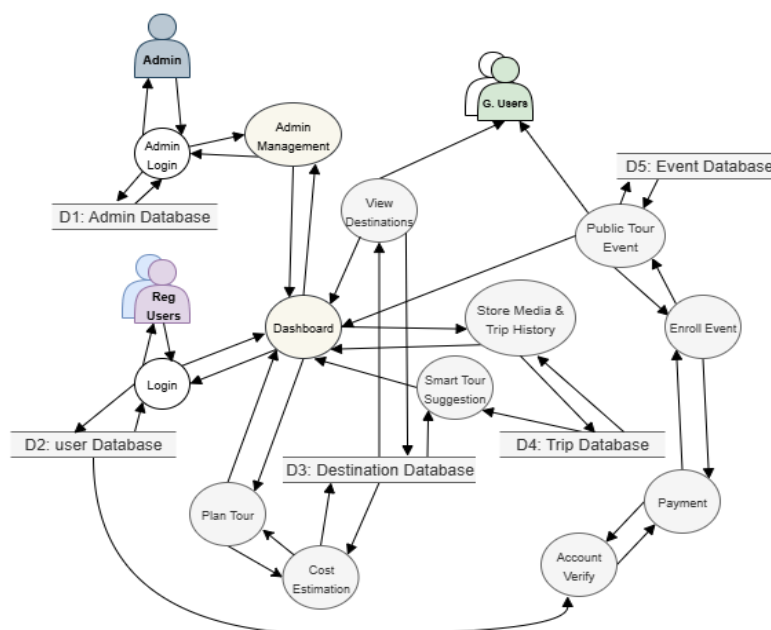


Figure 10. DFD Diagram

4.2 UI / Wireframe Design

Figma Link: <https://www.figma.com/design/PTCSILWMebFtyzluruazdb/Trip-Diary-Travel-Management--Tour-Suggestion-System?node-id=0-1&t=yHgO7Ag2TBh3MUR2-1>

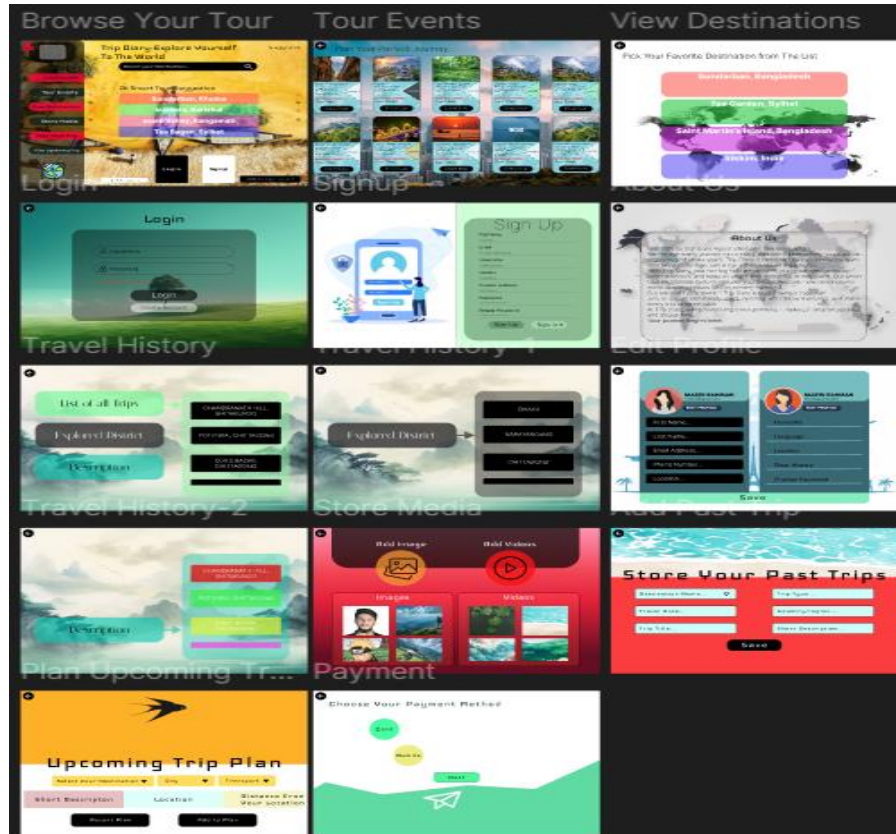


Figure 11. Initial Version Frames

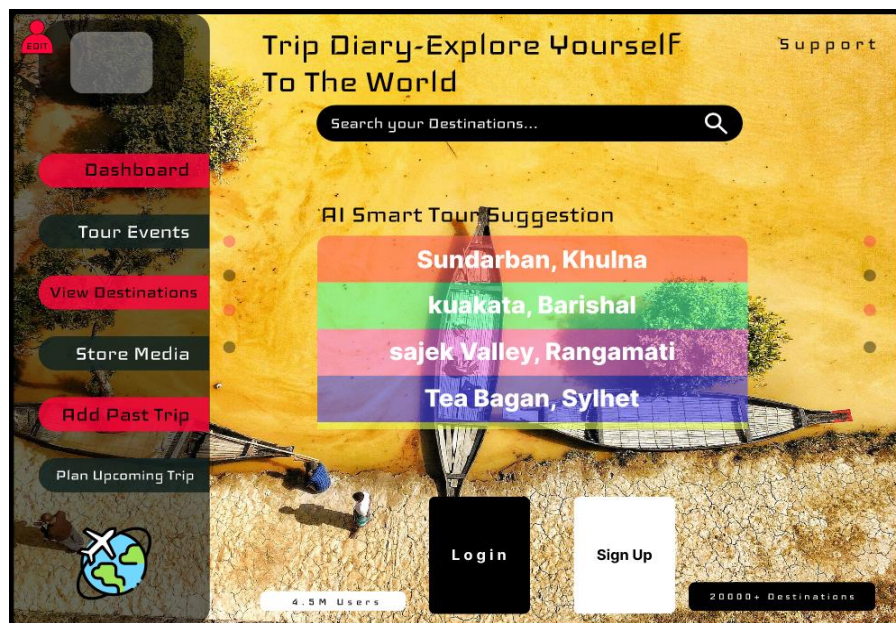


Figure 12. Dashboard

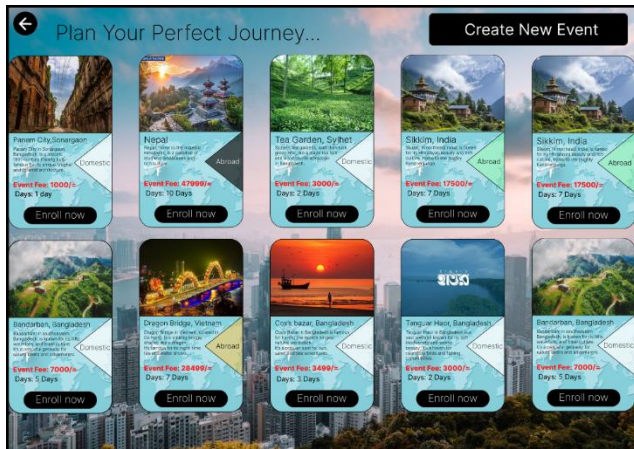


Figure 13. Tour Event

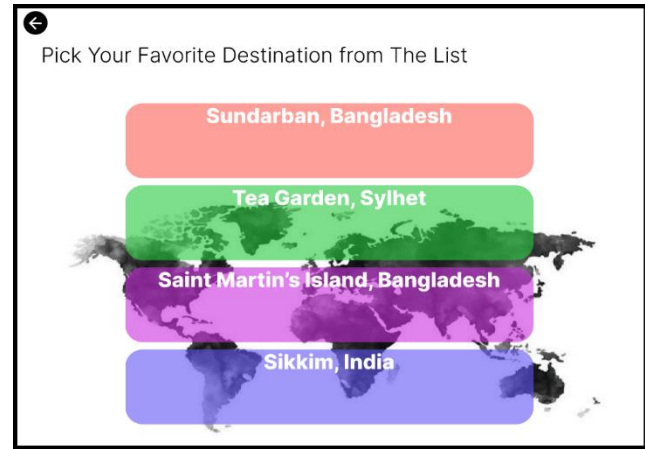


Figure 14. Destinations

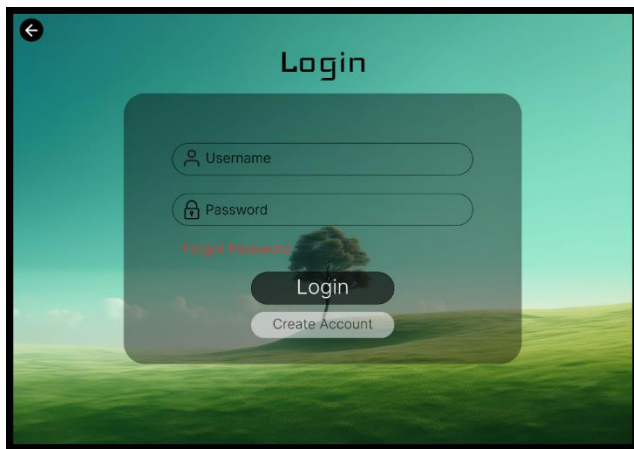


Figure 15. Login Form

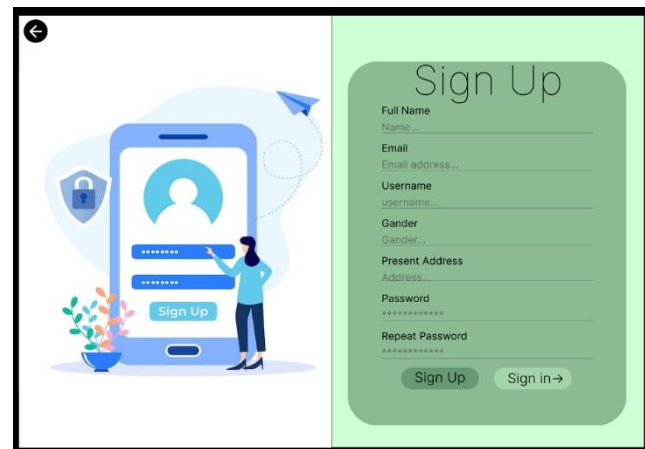


Figure 16. Signup Form

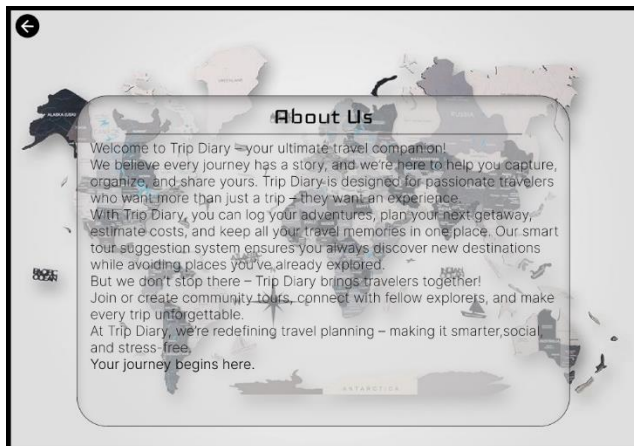


Figure 17. About us



Figure 18. Travel History-1



Figure 19. Travel History-2



Figure 20. Travel History-3



Figure 21. Edit Profile

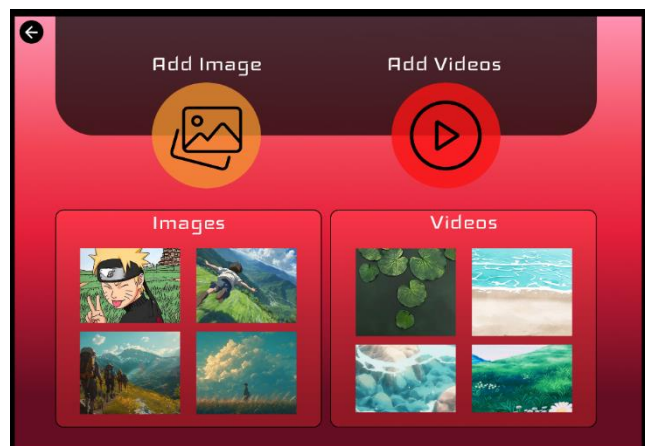


Figure 22. Store Media

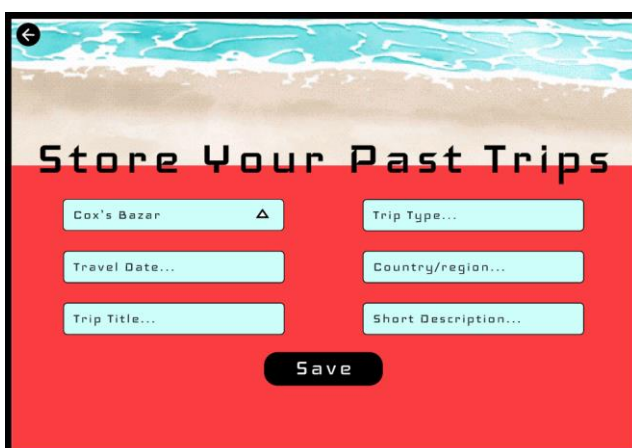


Figure 23. Past Trips

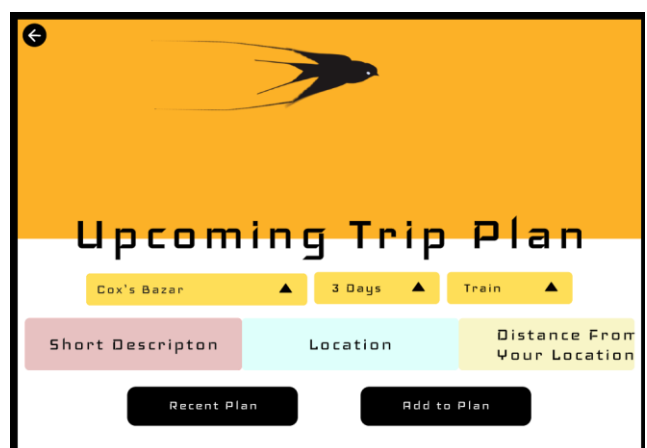


Figure 24. Upcoming Tour Plan

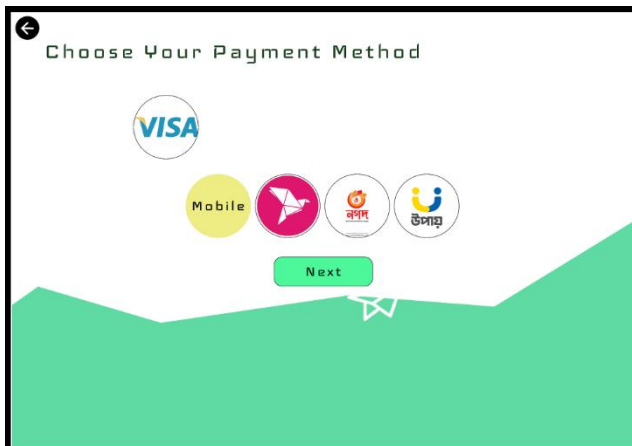


Figure 25. Payment/ Enrollment

5. GIT WORKFLOW

GitHub Link: <https://github.com/eyamin-000/TripDiary>

6. TESTING

Table 04. TC_01

Project Name: Trip Diary-Travel management & tour Suggestion System			Test Designed by: Eyamin Khan Emon	
Test case ID: TC_01			Test Designed Date: 05/10/2025	
Test Priority (Low, Medium, High): High			Test Executed by: Eyamin Khan Emon	
Module Name: Signup Page			Test Execution Date: 05/10/2025	
Test Title: Verify that a new user can successfully register with valid input data.				
Description: Make sure that a new user can easily set up an account by filling in all the necessary information accurately, and that the system successfully saves their data.				
Precondition: 1. The user is not already registered. 2. Internet connection and system server are active.				
Dependencies: The Signup page UI must be functional.				
Test Steps:	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
Open the Signup page	N/A	Signup form should load correctly	As expected	Pass
Enter Full Name	Emon	Name field accepts text input	As expected	Pass

Enter a valid Email address	emon@gmail.com	Email field accepts text input	As expected	Pass
Enter Username	emonkhan	Username field accepts text input	As expected	Pass
Enter Gender	male	Gender field accepts text input	As expected	Pass
Enter Present Address	sonargaon	Address field accepts text input	As expected	Pass
Enter Password & Re-enter password	emon01 emon01	Both Pass match successfully	As expected	Pass
Click “Sign Up”	-	User account should be created	As expected	Pass
Try “Login” with the new created account	Email: emon@gmail.com Pass: emon01	Login should Succeed	As expected	Pass

Table 05. TC_02

Project Name: Trip Diary-Travel management & tour Suggestion System		Test Designed by: Eyamin Khan Emon		
Test case ID: TC_02		Test Designed Date: 05/10/2025		
Test Priority (Low, Medium, High): Medium		Test Executed by: Eyamin Khan Emon		
Module Name: Signup Page		Test Execution Date: 05/10/2025		
Test Title: Verify that the system shows an error message when an invalid email address is entered during signup.				
Description: To make sure the system properly checks the email address format and stops the user from signing up if the email entered is not valid.				
Precondition: 1. The user is not already registered. 2. Internet connection and system server are active. 3. The user is not logged in.				
Dependencies: Email validation function must be implemented.				
Test Steps:	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
Open the Signup page	N/A	Signup form should load correctly	As expected	Pass
Enter Full Name	Emon	Name field accepts text input	As expected	Pass
Enter a valid Email address	Emon.gmail.com	System should detect invalid email format	As expected	Pass
Enter Username	emonkhan	Username field accepts text input	As expected	Pass
Enter Gender	male	Gender field accepts text input	As expected	Pass
Enter Present Address	sonargaon	Address field accepts text input	As expected	Pass

Enter Password & Re-enter password	emon01 emon01	Both Pass match successfully	As expected	Pass
Click “Sign Up”	-	User account should be created	As expected	Pass
Try “Login” with the new created account	Email: emon@gmail.com Pass: emon01	Login should Succeed	As expected	Pass
Post Condition: User Account is not Created.				

Table 06. TC_03

Project Name: Trip Diary-Travel management & tour Suggestion System		Test Designed by: Fahmida Sultana		
Test Case ID: TC_3		Test Designed date: 08/10/2025		
Test Priority (Low, Medium, High): Medium		Test Executed by: Fahmida Sultana		
Module Name: Login Page		Test Execution date: 09/10/2025		
Test Title: Verify login with valid username and password				
Description: Ensure that a registered user can successfully log in with the correct username and password				
Precondition: User already has an account				
Dependencies: Figma design link must be accessible				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open the login page 2. Enter username 3. Enter password 4. Click the “login” button	Username: fahmi73 Password: 777	User dashboard should log in successfully	As expected	Pass

Table 07. TC_04

Project Name: Trip Diary-Travel management & tour Suggestion System	Test Designed by: Fahmida Sultana
Test Case ID: TC_4	Test Designed date: 08/10/2025
Test Priority (Low, Medium, High): High	Test Executed by: Fahmida Sultana
Module Name: Login	Test Execution date: 08/10/2025
Test Title: Verify that user cannot login with invalid username and password	
Description: Ensure that the system does not allow login when incorrect username or password is entered	
Precondition: User account already exists in the system	

Dependencies: Login page accessibility required				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open login page 2. Enter invalid username 3. Enter invalid password 4. Click submit	Username: fam99 Password: 778	System should show an "login failed" message	As expected	Pass

Table 08. TC_05

Project Name: Trip Diary-Travel management & tour Suggestion System			Test Designed by: Eyamin Khan Emon	
Test case ID: TC_05			Test Designed Date: 05/10/2025	
Test Priority (Low, Medium, High): High			Test Executed by: Eyamin Khan Emon	
Module Name: Create Event			Test Execution Date: 05/10/2025	
Test Title: Verify that a registered user can successfully create a new public tour event.				
Description: Make sure that a registered user can create a new public tour event by entering all the required details correctly and that the event is saved and displayed for admin approval.				
Precondition: 3. The user must be logged in. 4. The “Create Event” page must be accessible. 5. Destination data should exist in the system database.				
Dependencies: 1. Event creation API and database connection must be functional. 2. Admin approval system should be active.				
Test Steps:	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
Click on “Create Event”	N/A	Signup form should load correctly	As expected	Pass
Enter event title	Title: “Cox’s Bazar Beach Tour 2025”	Title field accepts text input	As expected	Pass
Select destination	Destination: Cox’s Bazar	Destination field accepts text input	As expected	Pass
Enter event date and duration	Date: 15/12/2025 Duration: 3 Days	Date & duration field accepts text input	As expected	Pass
Enter cost per person	Cost: 5000 BDT	Cost field accepts text input	As expected	Pass
Add event description	Description: “A 3-day beach trip including sightseeing and group dinner.”	Description field accepts text input	As expected	Pass

Select event visibility	Option: Public	Option is selected	As expected	Pass
Click on “Create Event” button	-	System saves the event and shows message	As expected	Pass
Verify event in user profile	-	Newly created event appears under My Created Events with status Pending Approval	As expected	Pass
Post Condition: The new event is stored in the database with status “Pending Admin Approval.”				

Table 09. TC_06

Project Name: Trip Diary-Travel management & tour Suggestion System			Test Designed by: Eyamin Khan Emon	
Test case ID: TC_06			Test Designed Date: 05/10/2025	
Test Priority (Low, Medium, High): High			Test Executed by: Eyamin Khan Emon	
Module Name: Enrollment/Payment			Test Execution Date: 05/10/2025	
Test Title: Verify that a user can successfully complete enrollment payment for a public tour event.				
Description: Make sure that users can successfully pay the required enrollment fee to join a public tour event and that the transaction is processed securely and recorded in the system.				
Precondition: 1. The user must be logged in with a valid account. 2. A public tour event must be available and open for enrollment. 3. The user must have a valid payment method (e.g., card, mobile banking).				
Dependencies: Payment gateway and event enrollment APIs must be functional.				
Test Steps:	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
Navigate to the event details page	Event Id: 01	Event information should display properly	As expected	Pass
Click on “Enroll” button	-	System redirects to the payment page	As expected	Pass
Choose a payment method	Card	Payment option selection is accepted	As expected	Pass
Enter payment details	Card: 1111 1111 1111	System validates payment fields	As expected	Pass
Confirm the payment	Click “pay”	Payment is processed successfully	As expected	Pass
Verify enrollment confirmation	-	System shows success message	As expected	Pass
Check enrollment status in profile	Enrolled Events	Event should be enrolled	As expected	Pass

Table 010. TC_07

Project Name: Trip Diary-Travel management & tour Suggestion System			Test Designed by: Eyamin Khan Emon	
Test case ID: TC_07			Test Designed Date: 05/10/2025	
Test Priority (Low, Medium, High): High			Test Executed by: Eyamin Khan Emon	
Module Name: Enrollment/Payment			Test Execution Date: 05/10/2025	
Test Title: Verify that the system displays an error message when an invalid card number is entered during enrollment payment.				
Description: Make sure that the system properly validates payment information and prevents payment completion when the user provides an incorrect or invalid card number.				
Precondition: 1. The user must be logged in with a valid account. 2. A public tour event must be available and open for enrollment. 3. The user must have a valid payment method (e.g., card, mobile banking).				
Dependencies: Payment validation and error handling functions must be active.				
Test Steps:	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
Navigate to the event details page	Event Id: 01	Event information should display properly	As expected	Pass
Click on “Enroll” button	-	System redirects to the payment page	As expected	Pass
Choose a payment method	Card	Payment option selection is accepted	As expected	Pass
Enter payment details	Invalid card no: 1111 1111 11	System validates payment fields	As expected	Pass
Confirm the payment	Click “pay”	Payment is processed successfully	As expected	Pass
Verify enrollment confirmation	-	System shows success message	As expected	Pass
Check enrollment status in profile	Enrolled Events	Event should be enrolled	As expected	Pass
Postcondition: Payment is not processed, and no amount is deducted.				

Table 11. TC_08

Project Name: Trip Diary-Travel management & tour Suggestion System		Test Designed by: Fahmida Sultana	
Test Case ID: TC_8		Test Designed date: 09/10/2025	
Test Priority (Low, Medium, High): Medium		Test Executed by: Name	
Module Name: Edit Profile		Test Execution date: 09/10/2025	
Test Title: Verify that user can update profile information			
Description: Test the website edit profile page			

Precondition: User is logged in				
Dependencies: Profile page must be accessible through navigation from the dashboard				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Navigate to the edit profile page 2. Click the “Edit profile” button 3. Modify profile information 4. Click the “Save” button	New Email: koli@gmail.com Location: Comilla	A success message should appear confirming that the profile has been updated	As expected	Pass

Table 12. TC_09

Project Name: Trip Diary-Travel management & tour Suggestion System		Test Designed by: Fahmida Sultana			
Test Case ID: TC_9		Test Designed date: 08/10/2025			
Test Priority (Low, Medium, High): Medium		Test Executed by: Fahmida Sultana			
Module Name: Travel History		Test Execution date: 08/10/2025			
Test Title: Verify that the user can view and add travel history					
Description: Test that the user can view previous travel records and add new trips					
Precondition: User is logged in					
Dependencies: Travel History page must be linked correctly from the navigation menu in the prototype					
Test Steps		Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open the Travel History page 2. Click the “Add Image” or “Add Video” button 3. Upload a file 4. Click the “Save” button		Image: sundarban_trip.jpg	The new trip record should be saved and displayed in the list	As expected	Pass

Table 13. TC_10

Project Name: Trip Diary-Travel management & tour Suggestion System		Test Designed by: Marin Rahman		
Test Case ID: TC_10		Test Designed date: 9/10/2025		
Test Priority (Low, Medium, High): Medium		Test Executed by: Marin Rahman		
Module Name: Store Media		Test Execution date: 9/10/2025		
Test Title: Verify that user can add image or video in store media section				
Description: Ensure ‘‘Add Image’’ and ‘‘Add Video’’ buttons are visible				
Precondition: Store media frame open in prototype				
Dependencies: Figma prototype link accessible				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open the store media page in prototype mode 2. Check the visibility of ‘‘Add Image’’ and ‘‘Add Video’’ buttons 3. Click the ‘‘Add Image’’ button 4. Click the ‘‘Add Video’’ button	Select image file : beach.png Select video file : trip.mp4	Image and videos should be displayed in the media gallery under Store Media	As expected	Pass

Table 14. TC_11

Project Name: Trip Diary-Travel management & tour Suggestion System			Test Designed by: Marin Rahman	
Test Case ID: TC_11			Test Designed date: 9/10/2025	
Test Priority (Low, Medium, High): Medium			Test Executed by: Marin Rahman	
Module Name: Upcoming Trip Plan			Test Execution date: 9/10/2025	
Test Title: Verify that system shows error when unsupported file type is uploaded				
Description: Ensure that the system does not accept invalid or unsupported media file formats				
Precondition: User is on the store media page				
Dependencies: Media upload functionality must support validation				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. Open Store Media page 2. Click “Add Image” button 3. Try to upload file	File: document.pdf	System should show an error message: “Invalid file format” and prevent upload	As expected	Pass
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Table 15. TC_12

Project Name: Trip Diary-Travel management & tour Suggestion System		Test Designed by: Marin Rahman		
Test Case ID: TC_12		Test Designed date: 9/10/2025		
Test Priority (Low, Medium, High): Medium		Test Executed by: Marin Rahman		
Module Name: Upcoming Trip Plan		Test Execution date: 9/10/2025		
Test Title: Verify that user can create upcoming trip plan				
Description: Ensure destination , data, and transport dropdown accept user input and generate trip plan				
Precondition: User logged in , upcoming Trip page open				
Dependencies: Dropdown menus linked to sample options				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open upcoming trip plan page 2. Select destination from dropdown 3. Select day from dropdown 4. Select transport type 5. Click ‘‘Show plan’’ button	Destination: Comilla Day: Friday Transport: Train	Trip plan details should be generated and displayed on screen	As expected	Pass

Table 16. TC_13

Project Name: Trip Diary-Travel management & tour Suggestion System	Test Designed by: Marin Rahman
Test Case ID: TC_13	Test Designed date: 9/10/2025
Test Priority (Low, Medium, High): Medium	Test Executed by: Marin Rahman
Module Name: Upcoming Trip Plan	Test Execution date: 9/10/2025
Test Title: Verify system behavior when user does not select required fields	
Description: Test that the user cannot proceed without selecting destination or transport type	
Precondition: User logged in, Upcoming trip plan page	
Dependencies: Dropdown menus available	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open upcoming trip plan page 2. Leave destination and transport type unselected 3. Click "Show Plan" button	Destination : Blank Transport: Blank	System should show an error message: "Please select destination and transport type." No plan should be generated.	As expected	Pass

Table 17. TC_14

Project Name: Trip Diary-Travel management & tour Suggestion System			Test Designed by: Rashed Faruque	
Test Case ID: TC_14			Test Designed date: 09/10/2025	
Test Priority (Low, Medium, High): High			Test Executed by: Rashed Faruque	
Module Name: View Destinations			Test Execution date: 09/10/2025	
Test Title: Verify that destinations are displayed correctly on the View Destination page				
Description: Ensure that all destinations stored in the database are correctly fetched and displayed on the screen.				
Precondition: Database contains destination data.				
Dependencies: Database connection and page loading must be functional				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open the “View Destinations” page	-	Page loads successfully	As expected	Pass
2. Observe the destination list	Stored destinations	All destinations are displayed with correct names, regions and countries.	As expected	Pass

Table 18. TC_15

Project Name: Trip Diary-Travel management & tour Suggestion System			Test Designed by: Rashed Faruque	
Test Case ID: TC_15			Test Designed date: 09/10/2025	
Test Priority (Low, Medium, High): Medium			Test Executed by: Rashed Faruque	
Module Name: View Destinations			Test Execution date: 09/10/2025	
Test Title: Verify search functionality for destinations				

Description: Ensure that the search bar filters destinations based on the entered keyword.				
Precondition: Destination list is loaded				
Dependencies: Search feature and data filtering must be active				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open the “View Destinations” page	-	Page loads successfully	As expected	Pass
2. Type a destination name in the search box and press Enter	“Sundarban”	Only “Sundarban, Bangladesh” is displayed in the results	As expected	Pass

Table 19. TC_16

Project Name: Trip Diary-Travel management & tour Suggestion System			Test Designed by: Rashed Faruque	
Test Case ID: TC_16			Test Designed date: 09/10/2025	
Test Priority (Low, Medium, High): High			Test Executed by: Rashed Faruque	
Module Name: Add Past Trip			Test Execution date: /10/2025	
Test Title: Verify that user can successfully add a past trip with valid data				
Description: Ensure that the system saves trip details correctly when valid data is entered.				
Precondition: User is logged in, and Add Past Trip page is accessible				
Dependencies: Database and save functionality must be available.				
Test Steps	Test Data	Test Steps	Test Data	Test Steps
1. Open “Add Past Trip” page	-	Open “Add Past Trip” page	-	Open “Add Past Trip” page
2. Fill all fields with valid data and click Save	Destination: Sundarban; Date: 12/09/2025; Type: Adventure; Country: Bangladesh; Title: Sundarban Adventure Description: Jungle boat trip	Fill all fields with valid data and click Save	Destination: Sundarban; Date: 12/09/2025; Type: Adventure; Country: Bangladesh; Title: Sundarban Adventure Description: Jungle boat trip	Fill all fields with valid data and click Save

Table 20. TC_17

Project Name: Trip Diary-Travel management & tour Suggestion System			Test Designed by: Rashed Faruque			
Test Case ID: TC_17			Test Designed date: 09/10/2025			
Test Priority (Low, Medium, High): High			Test Executed by: Rashed Faruque			
Module Name: Add Past Trip			Test Execution date: 09/10/2025			
Test Title: Verify that user cannot save a past trip with missing mandatory fields						
Description: Ensure that the system shows an error message when required fields are empty.						
Precondition: Add Past Trip page is open.						
Dependencies: Form validation must be active.						
Test Steps		Test Data	Test Steps		Test Data	Test Steps
1. Open “Add Past Trip” page		-	Open “Add Past Trip” page		-	Open “Add Past Trip” page
2. Leave the “Trip Type” and “Travel Date” fields empty, then click Save		Destination: Sundarban; Country: Bangladesh; Title: Sundarban Adventure; Description: Jungle boat trip	Leave the “Trip Type” and “Travel Date” fields empty, then click Save		Destination: Sundarban; Country: Bangladesh; Title: Sundarban Adventure; Description: Jungle boat trip	Leave the “Trip Type” and “Travel Date” fields empty, then click Save

7. SOFTWARE PRODUCT METRICS

7.1 Function Point Calculation for Trip Diary

Based on the identified features, user stories, and system modules (Guest User, Registered User, and Admin), the following function points are identified:

External Inputs (EI)

- User signup form: 1
- Login authentication form: 1

- Add past trip form (destination, date, media upload): 1
- Planning upcoming trip form: 1
- Create public tour event form: 1
- Enroll in tour event (payment form): 1
- Profile update form: 1
- Cost estimation input form: 1
- Admin event approval form: 1
- Admin user management (block/delete): 1
- Admin destination management form: 1
- Media upload form (photos/videos): 1

Total External Inputs (EI): 12

External Outputs (EO)

- User dashboard display: 1
- Admin dashboard (system logs, stats): 1
- Travel history display: 1
- Upcoming trip planner view: 1
- Smart travel suggestions: 1
- Event list display (public tours): 1
- Enrollment confirmation / payment receipt: 1
- Cost estimation result page: 1
- Admin event report (approved/rejected events): 1

Total External Outputs (EO): 09

External Inquiries (EQ)

- Search destinations or events: 1
- View event details: 1
- Check enrollment/payment status: 1
- View user profile information: 1
- Retrieve travel suggestions based on history: 1

Total External Inquiries (EQ): 5

Internal Logical Files (ILF)

- User accounts database: 1
- Trip records (past and upcoming): 1
- Events database: 1

- Media files (photos/videos): 1
- Travel suggestions database: 1
- Destination database: 1

Total Internal Logical Files (ILF): 6

External Interface Files (EIF)

- Payment gateway integration: 1
- Email/notification service (for confirmations): 1
- Map API (for displaying locations and routes): 1

Total External Interface Files (EIF): 3

Function Point Calculation Table

Table 21. Function Point Table

Component	Count	Simple	Average	Complex	Weighted Score
EI	12	$5 \times 3 = 15$	$4 \times 4 = 16$	$3 \times 6 = 18$	49
EO	09	$3 \times 4 = 12$	$4 \times 5 = 20$	$2 \times 7 = 14$	46
EQ	5	$3 \times 3 = 9$	$2 \times 4 = 8$	$0 \times 6 = 0$	17
ILF	6	$3 \times 7 = 21$	$2 \times 10 = 20$	$1 \times 15 = 15$	56
EIF	3	$1 \times 5 = 5$	$2 \times 7 = 14$	$0 \times 10 = 0$	19

7.2 Object-Oriented & Class Metrics for Trip Diary

7.2.1 Weighted Methods per Class (WMC)

- User: 10
- RegisteredUser: 8
- Admin: 8
- Trip: 9
- Destination: 6
- Event: 10
- Enrollment: 5
- RecommendationEngine: 7
- RecommendationModel: 9
- CostEstimator: 6
- Payment: 7
- Media: 5

- AuthService: 8
- Notification: 4
- Location: 3
- Dashboard: 6

Total methods (sum): 109

Number of classes: 16

Average WMC = $109 / 16 \approx 6.81$

7.2.2 Depth of Inheritance Tree (DIT)

Estimated inheritance depths (examples):

- guest User to Register User: DIT for that branch = 2
- User to Admin: DIT = 2
- Payment-Card Payment-Third party Adapter: DIT = 3

From the system:

- Maximum DIT = 3
- Estimated DIT values across classes (approximate average): Average DIT= 2.33

7.2.3 Number of Children (NOC)

Representative child counts:

- User base class: NOC = 3
- Payment base class: NOC = 2
- Event base class: NOC = 1
- Other base classes have few or no children.

If we count four notable parents (User, Payment, Event, BaseService) with total children = 8

Average NOC = $8 / 4 = 2.0$

7.2.4 Coupling Between Objects (CBO)

Estimated coupling per subsystem:

- User management subsystem (User, AuthService, Notification, Profile, Dashboard): CBO= 5
- Trip management (Trip, Destination, Media, Location, RecommendationEngine): CBO= 6

- Event management (Event, Enrollment, Payment, Notification, Admin): CBO = 5
- Payment processing (Payment, ExternalGateway, Accounting, Notification): CBO = 7
- Recommendation system (RecommendationEngine, RecommendationModel, Trip DB, Destination DB, User DB): CBO = 5
- Media & Storage (Media, Storage API, Trip): CBO = 3

Sum of representative CBOs = 5 + 6 + 5 + 7 + 5 + 3 = 31

Number of subsystems used = 6

Average CBO = 31 / 6 = 5.17

7.2.5 Lack of Cohesion of Methods (LCOM)

- High cohesion classes (LCOM 0–2): 10 classes
- Medium cohesion (LCOM 3–5): 5 classes
- Low cohesion (LCOM 6+): 1 class

Assign representative LCOM values for averaging:

- High group average LCOM = 1.0
- Medium group average LCOM = 4.0
- Low group LCOM = 6.0

Computing average LCOM: $(10 \times 1.0 + 5 \times 4.0 + 1 \times 6.0) / 16 = (10 + 20 + 6) / 16 = 36 / 16 = 2.25$

7.2.6 Operation-Oriented Metrics for Trip Diary

1. Average Operation Size (LOC per method)

Assumed distribution:

- Simple operations: 50% of methods: average 4 LOC each
- Business logic operations: 35%: average 18 LOC each
- Complex operations: 15%: average 30 LOC each

Computing weighted average LOC per method: $0.5 \times 4 + 0.35 \times 18 + 0.15 \times 30 = 2 + 6.3 + 4.5 = 12.8$ LOC per method

2. Operation Complexity (Cyclomatic Complexity distribution)

Assumed distribution:

- Simple (CC 1–3): 65% of operations: mean CC = 2.0
- Medium (CC 4–7): 25%: mean CC = 5.5

- High (CC 8+): 10%: mean CC = 9.0

Weighted average CC: $0.65 \times 2.0 + 0.25 \times 5.5 + 0.10 \times 9.0 = 1.30 + 1.375 + 0.90 = 3.575 = 3.58$

3. Parameters per Operation

Assumed distribution:

- No parameters: 30%
- 1–2 parameters: 50% (mean = 1.5)
- 3–4 parameters: 15% (mean = 3.5)
- 5+ parameters: 5% (mean = 6)

Average parameters:

$0.3 \times 0 + 0.5 \times 1.5 + 0.15 \times 3.5 + 0.05 \times 6 = 0 + 0.75 + 0.525 + 0.30 = 1.575 = 1.58$ parameters / operation

7.3 Maintenance Metrics

7.3.1 Software Maturity Index (SMI)

Based on development phases and changes:

Version Analysis

Version 1.0 (Initial Release):

- Total modules (Mt): 23
- Modules changed (Fc): 0
- Modules added (Fa): 23
- Modules deleted (Fd): 0

Version 1.1 (First Update):

- Total modules (Mt): 27
- Modules changed (Fc): 5
- Modules added (Fa): 4
- Modules deleted (Fd): 0

Version 1.2 (Bug Fixes):

- Total modules (Mt): 28
- Modules changed (Fc): 6
- Modules added (Fa): 1

- Modules deleted (Fd): 0

SMI Calculations, $SMI = [Mt - (Fa + Fc + Fd)] / Mt$

- **SMI (Version 1.0) = $[23 - (23 + 0 + 0)] / 23 = 0.0$ (For Initial Release)**
- **SMI (Version 1.1) = $[27 - (4 + 5 + 0)] / 27 = 0.67$**
- **SMI (Version 1.2) = $[28 - (1 + 6 + 0)] / 28 = 0.75$**

Current SMI: 0.75 (75% stability)

This shows that the software is quite mature, with around 75% of the system staying stable across different versions.

8. Conclusion & Future Work

8.1 Conclusion

The Trip Diary project is a good example of a travel management and planning platform that brings together personal travel journaling, smart recommendations, and community events sharing all in one place. With features like secure user management, trip logging, a clever suggestion system, and event planning, it offers a digital solution for today's travelers.

Users can easily document their past adventures with photos and notes, plan future trips easily, and connect with others through shared events. The Smart Tour Suggestion System personalizes the experience by recommending new destinations based on a user's travel history and preferences, making it more engaging and user-friendly.

On the admin side, the system provides robust control through the Admin Panel, which allows for the management of users, destinations, events, and media content, ensuring a safe and organized platform.

In summary, the Trip Diary system successfully fulfills its primary goal, to make travel planning smarter, more interactive, and enjoyable. It strikes a perfect balance between simplicity and powerful features, helping users not only document their journeys but also discover exciting new adventures.

8.2 Future Work

There are so many exciting ways it could grow and improve in the future:

1. **AI-Powered Travel Assistant:** Using advanced machine learning to automatically generate itineraries and customize travel plans based on what you like, the weather, and your budget.
2. **Social Networking Features:** It could allow users to follow, message, and share their travel experiences with friends or fellow adventurers, helping to create a lively and engaging social travel community.
3. **Offline Access:** Allowing users to access their saved trips and notes even when they're not connected to the internet.
4. **Live Location and Tracking:** Adding GPS-based tracking could improve real-time trip sharing and offer navigation help.
5. **Gamification and Rewards:** Introducing fun elements like badges, points, or levels could inspire users to travel more and share their adventures.

With these future improvements, Trip Diary can transform into a smart travel companion, changing the way people plan, manage, and enjoy their journeys.

Links:

Notion Link: https://www.notion.so/Project-Proposal-Trip-Diary-Travel-Management-Tour-Suggestion-System-26eb245fec348084ab2bcc170da47494?source=copy_link

Figma Link: <https://www.figma.com/design/PTCSILWMebFtyzIuruazdb/Trip-Diary-Travel-Management---Tour-Suggestion-System?node-id=0-1&t=yHgO7Ag2TBh3MUR2-1>

GitHub Link: <https://github.com/eyamin-000/TripDiary>

Trello Board Link:

<https://trello.com/invite/b/688ae11418bc16d84fced640/ATTI18333e2e0be3bb18bada984b02bc41bcF93DFB00/trip-diary>

END